A regenerative hydrogen cycle for post-release buoyancy compensation in heavy-lift airships: real-time condensation limits and hold-down operation

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Abstract

In-flight payload release from heavy-lift airships leaves a sudden buoyancy surplus. A regenerative cycle could absorb it by burning part of the lifting hydrogen with atmospheric oxygen and keeping the product water as ballast. Closed-form relations and dynamic simulations of this cycle on four airships show that the hydrogen burned is modest, 44.8 kg per tonne at 3.17 kg/s on a 60-t-payload airship (Flying Whales LCA60T), but retaining its water in real time demands 446 MW of heat rejection: the heat-rejection unit would need about 8–11 kW/kg, whereas the classes that condense at the cycle's temperature fall 3–64 times short. A hold-down device (propellers or a tether) restrains the vehicle while the burn follows the installed heat rejection, turning the requirement into a recovery time: 10.6 min for the tonne with 10 MW on any airship. Divisible cargo delivered in consecutive releases

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reuses the same equipment, and their pace sets the cargo: half of the payload that the fixed equipment leaves free arrives in 3.7 h of consecutive releases; the LCA60T delivers 36 t in 6 h of one-tonne releases. The release yields work, but regeneration needs outside electricity; in flight it is possible but small, sized by the hours of sun the flight leaves free. Real-time condensation alone is thus out of reach, but a hold-down device makes tonne-class releases feasible on large airships, keeping 91–99 % of the hydrogen that venting loses.

Keywords: airship, buoyancy control, hydrogen, payload release, water condensation, requirements analysis

1. Introduction

Heavy-lift logistics is turning back to airships because they can load and unload at unprepared sites, without runway or port infrastructure [1, 2]: cargo airships have been pursued since the Cargolifter CL 160 of 1996 [1], and the Flying Whales LCA60T is designed to deliver cargo to remote sites without that infrastructure [3]. A central operational problem is buoyancy management at payload release, already a hazardous phase when a heavy load is handled below a hovering airship [1]: when the load leaves the vehicle, the lift it balanced becomes an upward surplus, and compensating this sudden increase of buoyancy is a critical issue of airship flight mechanics [4].

The established remedies lose material at every delivery. Venting lifting gas or dropping consumable ballast removes the surplus, but the gas or ballast lost must be replenished on the ground [2]. Sliding ballast provides attitude authority but not net heaviness [5]; vacuum chambers would close the cycle, but conventional ones are too heavy [6]; compressing helium into onboard tanks (control of static heaviness, COSH) closes it with compressors and pressure vessels [7], as a hydrogen airship could also do with its own lifting gas; regenerative fuel cells have also been applied to airships, but to pressure maintenance rather than to the post-release imbalance [8].

Hydrogen-lifted airships have an alternative: burning part of the lifting hydrogen with atmospheric oxygen. The lost gas volume reduces buoyancy, and the product water, retained on board, adds weight. Each kilogram of hydrogen burned with its water retained therefore adds about 22.3 kg of heaviness, against 13.4 kg when the same hydrogen is vented; the difference is the water, most of whose mass is oxygen taken from the air. That water holds the burned hydrogen in about 1300 times less volume than the gas did, at

ambient pressure, and electrolysis can later recover it. Early airships already managed the lift that their fuel took away: they recovered water from the engine exhaust as ballast [9, 10], and the Graf Zeppelin carried part of its fuel as gas, Blau gas, in its gas volume next to the hydrogen [2, 9]. That recovery took several hours, whereas modern payload release demands it in a matter of seconds to minutes. Sizing the burn rate as if the water were retained the instant it forms hides the requirement that governs the cycle: to keep the water, the whole exhaust—mostly nitrogen—must be cooled below its dew point while the maneuver is under way. A hull of hundreds of thousands of cubic meters in the open air suggests that this would be easy, but as a passive exchanger the hull of the LCA60T would reject only 3.2–15.8 MW, against the 446 MW that a one-tonne release needs.

Every device of the cycle is treated here as a black box, defined by its input, output and key parameters, and the cycle is evaluated in two configurations. Alone, as the sole buoyancy control for a single release, the most demanding case, the cycle gives the burn rate, the heat rejection and the equipment mass that real-time compensation requires, and existing equipment classes show whether these fit within the payload of an airship. Coupled with a hold-down device, of the kind that airships already use to stay on the ground or to fly statically light, the cycle no longer has to keep pace with the release: the device restrains the vehicle and the burn follows the installed heat rejection. This second configuration covers single releases and divisible cargo delivered in consecutive releases, and shows which releases and airships the cycle can serve and at what cost against other methods.

The hydrogen burned turns out to be modest, and the burn rate follows in closed form for any airship in the ideal bound of instantaneous condensation. What limits the cycle is retaining the water in real time: the heat-rejection unit would need 3–64 times the power per kilogram of the equipment classes that condense at the cycle's temperature, so real-time condensation alone is out of reach. A hold-down device turns that requirement into a recovery time and makes tonne-class releases feasible on large airships; design relations give the recovery time, the largest release that fits the payload and the cargo delivered in consecutive releases. The cycle keeps most of the hydrogen that venting loses and the release yields work, but regeneration needs outside electricity; in flight it is possible but small, sized by the hours of sun that the flight leaves free. These conceptual-stage results identify the heat-rejection unit and the hold-down device as the subsystems that will drive the cycle's development.

Section 2 describes the cycle and its black boxes, and Section 3 the model. Section 4 sizes the burn in the ideal bound. Section 5 sets the requirements of real-time condensation against existing equipment and then couples the cycle with a hold-down device, for single releases and for delivery in consecutive releases. Section 6 covers the energy of the cycle and its regeneration, and Section 7 compares the cycle with other methods. The model is implemented in Python and verified at the component level, and the authors assess the credibility of its results with the NASA standard for models and simulations, NASA-STD-7009B [11] (Section 8 and supplementary material). Section 9 discusses the limits of the results, and Section 10 concludes.

2. System description

The cycle compensates a release by burning lifting hydrogen and keeping its water on board, instead of venting the gas, and it works in two stages (Figure 1). In the compensation stage, the energy conversion unit burns hydrogen from the gas cells with ambient air at the rate that the release demands, and the gas volume lost reduces the lift. The heat-rejection unit then condenses part of the water in the exhaust, which is kept as ballast, while the nitrogen and the uncondensed vapor leave the vehicle. The work of the burn powers the fans of the heat-rejection unit and the hold-down device.

The regeneration stage comes later: the regeneration unit electrolyzes the stored water back into hydrogen, and until then the water is kept in collapsible tanks as dense, unpressurized ballast. To prevent unintended buoyancy changes, the regenerated hydrogen is continuously compressed into high-pressure storage tanks, which also carry the make-up hydrogen for any water not retained and keep room for the hydrogen to be regenerated. These tanks act as an accumulator, feeding the gas cells through a pressure regulator whenever more lift is needed. The oxygen of the water is vented: it came from the ambient air during combustion, so venting it restores the vehicle to its initial mass, whereas retaining it would add 7.9 kg of ballast per kilogram of regenerated hydrogen. Overall, the cycle takes in air and electricity and gives off nitrogen, uncondensed vapor, oxygen and heat.

Since venting the oxygen lightens the vehicle, electrolysis should ideally run when more lift is wanted, such as on the ground or while the next payload is loaded. In free flight, that lift accrues as slowly as the regeneration and, unless it is wanted, has to be trimmed like the thermal drift of the gas (Section 9). Compressing lifting gas into the tanks trims it (Section 6);

keeping the oxygen on board would avoid it, at the cost of vessels of its own. Finally, the gas cells operate at ambient pressure, as their slight superpressure is negligible.

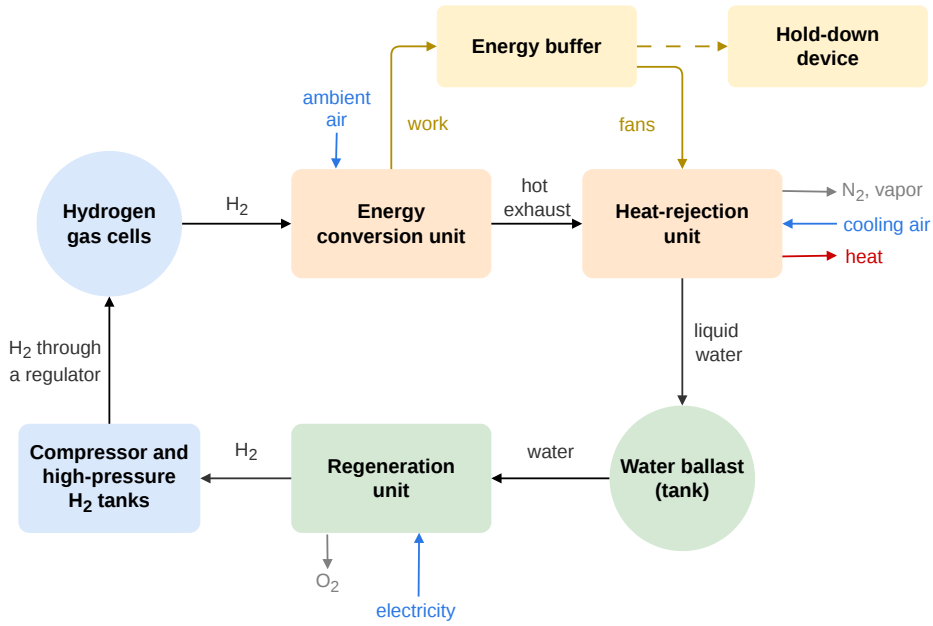


Figure 1: Regenerative hydrogen cycle for post-release buoyancy compensation.

Subsystems and requirements. Table 1 defines the cycle's subsystems as black boxes. The boxes are defined by performance requirements (key parameters), derived in Sections 4–5, not by hardware. Boundary temperatures for the thermal units are detailed in Section 5.1, and the conversion efficiency (η) accounts for any heat recovered as work.

Table 1: Black boxes of the cycle (LHV: lower heating value).

Black box	**Input**	**Output**	**Key parameters**
Energy conversion unit	H_2, ambient air	Work η LHV per kg H_2; hot exhaust	Burn rate $\dot{m}_{H_2}$; efficiency η
Heat-rejection unit	Exhaust, ambient air, electricity for the fans	Liquid water (fraction ϕ); N_2 and vapor to ambient	Installed power P_{cap}; response time τ; mass per kW; fan power fraction f
Energy buffer	Electricity	Electricity	Power and energy per kg
Water storage	Liquid water from the heat-rejection unit	Water to the regeneration unit	Tank mass per kg of water
Regeneration unit	Water, electricity	H_2 to the tanks; O_2 to ambient	Electricity per kg H_2
Hydrogen storage	Regenerated H_2; lifting gas to trim the lift; electricity	H_2 to the gas cells through a pressure regulator	H_2 stored per tonne released; compressor flow; electricity per kg compressed; regulator flow up to $\dot{m}^*_{H_2}$
Hull air system	Ambient air	Air into the hull as the gas cells shrink; out when hydrogen is returned	Volume flow $\dot{V} = \dot{m}_{H_2}/\rho_{H_2}$
Hold-down device	Onboard electricity	Downward force on the vehicle	Force; response time τ_{h} or rate of force $\dot{F}_{\text{h}}$; energy

Operation of the maneuver. Without a hold-down device, the heat-rejection unit is pre-activated (Section 3.4) and the burn begins 7.1 s before release (Section 4), so the vehicle sinks D_{lim} by the release and comes to rest at the new equilibrium, $(1 + s)\, D_{\text{lim}}$ below its initial altitude, from where the altitude controller slowly trims it back (Appendix B; Supplementary Section S7). Alternatively, a hold-down device (Section 5) can take the weight of the released load at once or ramp its force concurrently, so that the burn proceeds at the pace of the installed heat rejection.

3. Modeling

3.1. Vertical dynamics and inertia

The airship is modeled along the vertical axis [12, 13]:

$$M_I\,\dot{v} = \rho_{\text{air}}(h)\,g\,V_{\text{gas}} - M\,g - \tfrac{1}{2}\,C_D\,A\,\rho_{\text{air}}(h)\,v\,|v|, \qquad \dot{h} = v, \tag{1}$$

where $M = M_{\text{str}} + m_{\text{load}}(t) + m_{H_2}(t) + m_w(t)$ is the mass of the empty airship, payload, lifting gas and retained water, $V_{\text{gas}} = m_{H_2} R_{H_2} T_1 / P_{\text{atm}}$ the volume of the (ideal, isothermal) lifting gas, and A the drag reference area. The air density follows $\rho_{\text{air}}(h) = \rho_0\, e^{-(h-h_0)/H_s}$ with $H_s = 10.4$ km, the local density scale height of the standard atmosphere at the reference state [14, 15].

When the airship accelerates vertically, it moves more than its own mass: it carries the air that fills the hull volume freed by the burned hydrogen and sets in motion a mass of outside air proportional to the displaced volume. Its inertia M_I is therefore larger than its mass:

$$M_I = M + \rho_{\text{air}}\,(V_{\text{hull}} - V_{\text{gas}}) + k\,\rho_{\text{air}}\,V_{\text{hull}}, \tag{2}$$

with V_{hull} the total hull volume (gas cells and the air around them) and k the transverse added-mass coefficient of a prolate spheroid of fineness ratio ℓ/D in potential flow [16, 17]. The coefficient rises with the fineness ratio, from $k = 0.5$ for a sphere to $k \to 1$ for an infinite cylinder (Figure 2, right), and each airship uses its own (Table 2). At equilibrium the airship weighs as much as the air its gas displaces, so the inertia becomes $M_I = (1+k)\,\rho_{\text{air}} V_{\text{hull}}$ whatever the airship weighs and it returns to this value after every burn. Section 9 compares it with flight data.

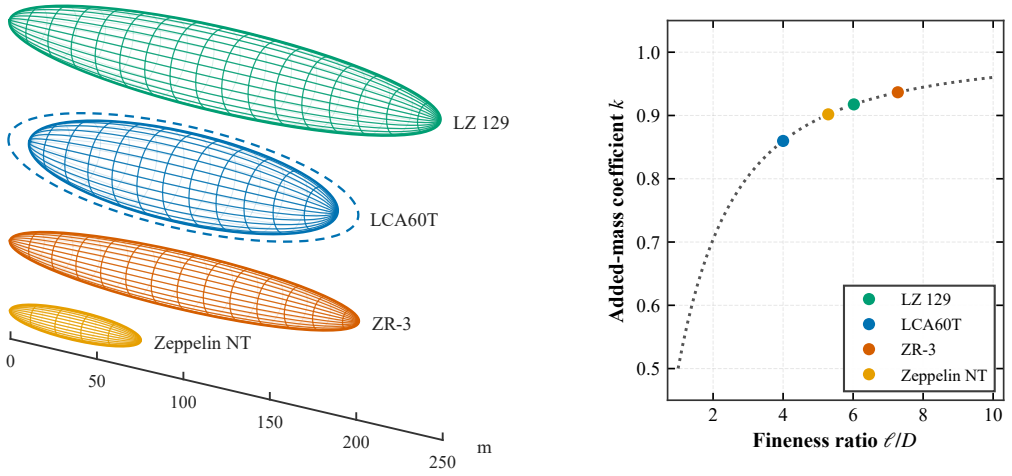


Figure 2: Left, hulls of the four airships as prolate spheroids with the published length and diameter, to scale in an orthographic view, with the mesh rings equally spaced on all hulls; these spheroids enclose 0.94–1.02 times the hull volume of Table 2. For the LCA60T, whose published helium volume is held inside a larger envelope, the mesh is the spheroid of the same fineness ratio that encloses only that volume (177 × 44 m), and the dashed outline is its published envelope. Right, added-mass coefficient k versus fineness ratio [16, 17].

The maneuver starts from rest ($v = 0$) at the equilibrium altitude $h_0 =$ 50 m, of the order of the lift height of ship-to-shore container cranes [18] and of the long lines, over 20 m, of helicopter external-load operations [19]; the results do not depend on it, since the air density is referenced to h_0. The payload is released as a step change in load, which represents an emergency drop or the instantaneous release of a hook under load. The water retained per kilogram of hydrogen burned is $r\phi$, where ϕ is the fraction of the product water retained, with $r = M_{H_2O}/M_{H_2} = 8.937$.

Table 2 lists the four airships evaluated, each defined by its published hull volume (LZ 129 and ZR-3 [9, 20, 21]; LCA60T [3, 22]; Zeppelin NT [23]), with k, A and M_I computed as in Section 3.1; the modern helium airships are evaluated as notional hydrogen variants with the same hull volume, empty mass and payload, which is conservative, since hydrogen lifts more than helium. The left of Figure 2 draws their hulls to scale.

Table 2: Airships evaluated.

Airship	V_{hull} **(m³)**	ℓ/D	k	M_{str} **(t)**	$m_{\text{load,max}}$ **(t)**	A **(m²)**	M_I **(t)**
LZ 129 Hindenburg	216,624	6.0	0.92	112.9	19.0	8030	508.8
Flying Whales LCA60T	180,000	4.0	0.86	100.0	60.0	7854	410.1
ZR-3 Los Angeles	79,287	7.3	0.94	39.2	5.0	4368	188.1
Zeppelin NT	8425	5.3	0.90	–[a]	1.0[b]	838	19.6

[a]Not published; the simulations take the hull full of hydrogen.
[b]Not published ("reserved" in the type certificate); the one-tonne reference release is used.

3.2. Altitude control and burn sizing

A proportional–derivative (PD) controller acting on the gas volume governs the altitude (Appendix B), with the derivative gain dynamically scheduled against the instantaneous inertia M_I to maintain a constant damping ratio ($\zeta = 1.30$). The controller outputs a bidirectional mass flow rate: negative commands trigger combustion, while positive commands re-inject stored gas when the vehicle sinks below its equilibrium, both saturated at the rate sized in Section 4. Simulating lift via local air density at constant gas volume stiffens this loop without altering peak excursions (Supplementary Section S7). The hold-down device subtracts a downward force in Equation (1), commanded to cancel the static imbalance and to correct the altitude through a second-order loop ($\omega_n = 0.3$ rad/s, damping ratio 1.0). This actuator features a first-order lag (1.0 s) and saturates at $(1+\mu)\, m_{\text{load}}\, g$. For anticipated releases, a feedforward ramp leads the release by one time constant; the largest lag that keeps both the rise and the sink within $h_{\max}$ is found iteratively.

3.3. Water retention and heat to reject

Burning hydrogen removes its buoyancy, eliminates its mass, and adds the retained water. The net heaviness gained per kilogram burned is:

$$L(\phi) = \frac{\rho_{\text{air}}}{\rho_{H_2}} - 1 + r\,\phi, \tag{3}$$

where ϕ is the fraction of product water retained. With full retention ($\phi = 1$), $L = 22.3$ kg per kg of hydrogen burned; thus, releasing one tonne requires

44.8 kg of hydrogen. If all vapor escapes ($\phi = 0$), $L = 13.4$. Using atmospheric instead of stored oxygen reduces the required hydrogen by 35.6 %, as the oxygen enters the vehicle as product water.

To retain water, the entire exhaust (34.3 kg of stoichiometric air per kg of hydrogen) must be cooled below its dew point. With enthalpies referenced to 25 °C, the rejected heat per kilogram of hydrogen is (Appendix C):

$$q(\eta, \phi) = (1 - \eta)\,\mathrm{LHV} + \phi\, r\, L_v + \Delta h_{\mathrm{sens}}, \tag{4}$$

where η is the LHV fraction exported as work, L_v is the latent heat at 25 °C, and Δh_{sens} is the sensible heat between the incoming outside air (30 °C design ambient) and the exhaust leaving at $T_{\mathrm{out}} = 40$ °C. The vapor is diluted in $N_{\mathrm{in}} = 1.89$ mol of inert gas per mol of water (35 % water by mole, dew point 73 °C at 1 atm). This dilution limits the condensable fraction via the saturation pressure p_{sat} [24]:

$$\phi_{\max}(T_{\mathrm{out}}, P) = 1 - N_{\mathrm{in}}\,\frac{p_{\mathrm{sat}}(T_{\mathrm{out}})}{P - p_{\mathrm{sat}}(T_{\mathrm{out}})}. \tag{5}$$

3.4. Heat-rejection unit: power, response and mass

The heat-rejection unit is modeled as a black box with installed power P_{cap} and a first-order thermal lag τ. Once started, its rejected power P_c rises as:

$$\tau\,\dot{P}_c = P_{\mathrm{cap}} - P_c. \tag{6}$$

It condenses water at the rate $\min(r\,\phi_{\max}\,\dot{m}_{H_2},\ \ P_c/h_w)$, where $h_w = q/(r\,\phi_{\max})$ is the rejected heat per kilogram of condensed water. The model lets the unit cool completely the part of the exhaust it handles, an optimistic bound, since real condensers in series must first remove the sensible heat above the dew point. To ensure the unit reaches full capacity before the exhaust arrives, the unit is activated 3 response times prior to the preconditioning burn (Supplementary Section S7); a unit following the burn from zero is kept only as a sensitivity case. Uncondensed vapor vents with the nitrogen. Ideal bounds assume instantaneous condensation ($\tau = 0$, $P_{\mathrm{cap}} \to \infty$); the burn rate is sized with full retention ($\phi = 1$) and the heat with the fraction 0.99 that the unit retains (Section 5.1); with that fraction, the rate would be 0.4 % higher, well inside the uncertainty of the vertical inertia, −9 to +14 % (Section 9).

Operationally, the unit is modeled for hover, with fans, and the exhaust leaves it at $T_{\mathrm{out}} = 40$ °C, where at 1 atm it retains a fraction 0.85 of its

water. To retain 0.99 of the water, as the heat of the ideal bound assumes, the exhaust is pressurized to 14 atm, which raises its dew point to 151 °C. The dynamics leave the mass of the hardware out; Section 5.1 derives the budget it may use.

3.5. Implementation and assumptions

The model is implemented in Python, integrating the equations of motion with SciPy's LSODA solver to handle stiff segments induced by the altitude controller. The model rests on four groups of assumptions:

- *Atmosphere and lifting gas:* The gas is ideal and isothermal, evaluated at standard sea-level conditions; the 30 °C design day enters only the thermal balance; in the dynamics it would lower the inertia by 5 % and raise the installed rate by 2.6 % (Supplementary Table S1). Intake air is assumed dry, which is conservative; at 50–90 % relative humidity, intake moisture would condense alongside product water, reducing the required hydrogen burn per tonne by 1.6–2.9 %.

- *Combustion:* Stoichiometric combustion is assumed. The stoichiometric assumption maximizes the vapor concentration, a favorable lower bound for sizing the heat-rejection unit; real combustors often use excess air [25], which dilutes the vapor.

- *Maneuver dynamics:* Payload release is modeled as an instantaneous step change in load. The altitude controller possesses perfect, instantaneous knowledge of the onboard mass and release timing. Aerodynamic drag is continuously modeled, but gusts are neglected, and the hull is assumed to admit air without delay as gas cells shrink.

- *Subsystem boundaries:* The hold-down device is treated as an ideal black box (e.g., ideal propellers) powered by the cycle's engine or, beyond its reach, by the airship's plant. Long-term processes like regeneration and hydrogen storage (which span hours to days) are sized via energy and mass balances rather than dynamic simulation. During the short maneuver, storage tanks act as ideal sources without transient regulator dynamics. Equipment masses are nominal, those of the lightest classes found (Section 5.1) and of the panels of Section 6, with no growth margin; Section 8 gives its effect.

Supplementary Table S1 summarizes all the assumptions, and Supplementary Table S2 ranks the phenomena.

4. Ideal bound: instantaneous condensation

Assuming instantaneous water retention, the sole requirement is the burn rate. To gain time, a continuous preconditioning burn at the installed rate $\dot{m}$ starts before the release (Appendix A). This phase lowers the vehicle by D_{lim} and pre-compensates a payload fraction α. Post-release, the ongoing burn opposes the upward force $m_{\mathrm{load}}\, g$. To achieve a "clean burn"—bringing the vehicle to rest exactly at the new equilibrium (m_{load}/L) without burning excess gas or requiring reinjection—the pre-release downward momentum is canceled, yielding:

$$\dot{m}^*_{H_2} = \sqrt{\frac{g\, m^3_{\mathrm{load}}}{6\, M_I\, L^2\, D_{\mathrm{lim}}\, (1+s)^3}}, \qquad \alpha^* = \frac{1}{1+s}, \qquad s = \sqrt{1 - \frac{m_{\mathrm{load}}}{M_I}}. \quad (7)$$

The maneuver ends at rest $(1+s)\, D_{\mathrm{lim}} \approx 2\, D_{\mathrm{lim}}$ below initial altitude, never rising above the release point. The clean burn therefore meets any precision threshold $h_{\max}$ larger than this depth. An operational $D_{\mathrm{lim}} = 0.10$ m is selected to consume only 20 % of the 1-m precision threshold, of the order of the set-down tolerance of the precision load-placement task flown with helicopter slung loads [26], preserving margins for timing and gusts (Supplementary Table S2). Since $s \approx 1$, preconditioning offsets half the payload ($\alpha^* = 0.5$) across all airships, scaling the rate as $m^{3/2}_{\mathrm{load}}/\sqrt{M_I}$. Because the burn rate bounds how fast the net force can change, the vehicle follows a jerk-limited motion: with $\ddot{h}_{\mathrm{s}} = m_{\mathrm{load}}\, g/M_I$ the acceleration that the release imposes and $j = L\, g\, \dot{m}^*_{H_2}/M_I$ the jerk of the burn, Equation (7) reads $D_{\mathrm{lim}} = \ddot{h}^3_{\mathrm{s}}/[6\, j^2\, (1+s)^3]$, the scaling of the jerk-limited point-to-point motions of motion control [27]. What the cycle adds is the clean condition, which fixes the preconditioning fraction, and the retained water, which sets L.

Table 3 summarizes the bounds for a one-tonne release. The clean burn, sized via Equation (7) with $D_{\mathrm{lim}} = 0.10$ m, satisfies all altitude thresholds. The reactive-control columns give the installed rate that a strictly reactive controller (Appendix B), with no pre-release burn, needs to hold the rise within the precision (≤ 1 m) or safety (≤ 5 m) threshold: 2.8–3.0 times the rate of the clean burn sized here ($D_{\mathrm{lim}} = 0.10$ m) for the 1-m threshold. Against a clean burn sized for the same excursion, it needs 4.2–4.4 times its rate at 5 m, 6.3–6.7 times at 1 m and 12–13 times at the $(1+s)\, D_{\mathrm{lim}}$ of the clean burn sized here (Supplementary Section S7). By compensating

the whole release in $t_{\text{burn}} = m_{\text{load}}/(L\,\dot{m}^*_{H_2})$ seconds (Table 3), this clean burn dictates the severe thermal requirements analyzed in Section 5. The net hydrogen consumed is the same in every mode and on every airship, a static requirement ($m_{\text{load}}/L = 44.8$ kg).

Table 3: Ideal bound (instantaneous condensation, all the water retained) for a one-tonne release; it does not depend on the exhaust pressure. $t_{\text{burn}} = m_{\text{load}}/(L\,\dot{m}^*_{H_2})$: duration of the whole clean burn, before and after the release.

	Clean burn				**Reactive control (kg/s)**	
Airship	$\dot{m}^*_{H_2}$ **(kg/s)**	t_{anticip} **(s)**	t_{burn} **(s)**	**Sink rate (m/s)**	**Rise ≤ 1 m**	**Rise ≤ 5 m**
LZ 129 Hindenburg	2.84	7.9	15.8	0.038	8.0	2.4
Flying Whales LCA60T	3.17	7.1	14.2	0.042	9.0	2.7
ZR-3 Los Angeles	4.68	4.8	9.6	0.063	13.7	4.1
Zeppelin NT	14.75	1.5	3.0	0.195	44.1	12.9

5. Real-time limitations and the hold-down architecture

5.1. Without a hold-down device: what the ideal bound demands and what exists

The burn rates of Table 3 assume that the water condenses as it forms. Condensing nearly all of it ($\phi = 0.99$) means rejecting $q = 141$ MJ per kilogram of hydrogen, so a one-tonne release on the LCA60T demands $P_{\text{rej}} = \dot{m}^*_{H_2}\,q = 446$ MW of heat rejection, continuously during the maneuver; passive cooling through the hull, 3.2–15.8 MW, falls far short. The same burn also empties the gas cells quickly: the hull must admit the air that replaces the burned hydrogen, 526 m^3 per tonne released, at up to 37 m^3/s on the LCA60T and 173 m^3/s on the Zeppelin NT, and let it out at the same rate when the controller returns hydrogen.

Two limits stand in the way: physics and hardware. The heat must be rejected in a low-temperature band. At 1 atm the exhaust condenses between its dew point and the exit of the heat-rejection unit (Section 3.3), where it retains only 0.85 of its water, and less on a warmer day (Table 4). Retaining 0.99 would need cooling the exhaust to -2 °C. As modeled, the cycle instead compresses it to 14 atm (Section 3.4), at the cost of at least 21 MW of compression work.

For the hardware, Table 5 gives the lightest class found for each black box. Three arrangements are evaluated: without a hold-down device, at the burn rate of the ideal bound (NHD), and with one (HD), with the exhaust at 14 atm or at 1 atm. All three carry every box of the table, except the hold-down device in NHD and the exhaust compressor at 1 atm. In all three the engine drives the fans, and the cycle can power its own cooling: it needs a conversion efficiency of only 0.15–0.19 at 1 atm and 0.21–0.25 at 14 atm with the fan power of aircraft radiators (Appendix C), below the 0.30–0.45 of hydrogen engines [28].

Table 4: Effect of the outside-air temperature at 1 atm.

Outside air (°C)	15	20	25	30	35	40
Exit of the heat-rejection unit (°C)	25	30	35	40	45	50
Water retained, $\phi_{\max}$	0.94	0.92	0.89	0.85	0.80	0.74
H_2 per tonne (kg)	46.0	46.4	46.9	47.7	48.7	50.1

Table 5: Black boxes and the lightest class found for each. ACC: air-cooled condenser; PEM: proton exchange membrane. The classes are mass references, not designs: the lightest engine is an aviation gasoline engine (Rotax 916 iS) and the heaviest a hydrogen genset (DEUTZ TCG 7.8 H_2); the regeneration unit takes the stack of aviation PEM fuel cells for the electrolyzer; the ACC is the power-plant system inventoried in [29]. The exhaust compressor is needed only at 14 atm. The tanks, the PEM stack and the hydrogen compressor are sized to regenerate in 24 h the hydrogen of a tonne released at 1 atm; at 14 atm, which retains more water, they take 26 h.

Black box	**Lightest class**	**Specific mass**
Energy conversion unit	H_2 engine, $\lambda = 1$ [30, 31]	26–126 t per kg/s
Heat-rejection unit	Power-plant ACC, bundles [29]	0.653 kW/kg
Exhaust compressor	Industrial gas turbine[a]	3.8–4.0 t per 10 MW
Hydrogen tanks	Type IV, 350 bar [32]	0.87–0.91 t per t released
Hydrogen compressor	Electric, catalog [33]	0.55 t
Regeneration unit	PEM stack [34]	0.04–0.18 t per t released
Water storage	Collapsible fabric tank [35]	1.3–2.7 kg per t released
Hold-down device	Propellers or tether	–

[a]Scaled by compression work from the 285-t package of an industrial gas turbine, generator included [36].

The energy closes but the mass does not. Even if the whole payload of the

LCA60T, less the tonne to be released, went to the cycle, what would remain for the boxes that grow with the burn rate weighs less than an engine alone (Supplementary Table S7). With a burner and a battery that drives the fans instead [37], the heat-rejection unit would still have to reject at least 8–11 kW per kilogram of its own mass at 1 atm; at 14 atm no unit fits, since the exhaust compressor alone would weigh 2.8 times the payload. The classes that condense at the temperature of the cycle, power-plant air-cooled condensers (ACC) [29] and the water-recovery equipment of rigid airships [9], fall 3–64 times short (Supplementary Table S8); only aircraft heat exchangers reach the requirement, and only the 2035 classes, whose heat-rejection temperature of 100–160 °C [38] lies far above the condensation band at 1 atm. Condensers that recover water from aero-engine exhaust, studied for the water-enhanced turbofan, are sized for the cold air of cruise, and their water cycle holds only down to about 3 km on a standard day [39]. In tonnes, with the exhaust at 14 atm, on the three larger airships without a hold-down device the boxes and the tonne weigh over ten times the payload of the LCA60T, and the equipment that only condenses weighs 10 times the engine (Section 5.2).

What fits in the payload does not hold the vehicle either. At the specific power of air-cooled condenser bundles (Table 5), only the 10-MW case of Figure 3 fits, with 15 t of bundles (74 t as a whole power-plant system) and only on the LCA60T; the 100-MW case would weigh more than the payload of every airship. Figure 3 simulates the four airships at 1 atm with the response of the heat-rejection unit, started early enough that its lag does not limit the maneuver. With unlimited power the simulations approach the ideal bound, but with 100 or 10 MW each airship rises 1.6–2.7 m, beyond the 1-m precision threshold, before returning to equilibrium. The missing ballast explains it: with 10 MW most of the water leaves with the nitrogen, and the controller burns about twice the hydrogen of the ideal bound without holding the vehicle (Supplementary Section S8). Real-time condensation therefore fails twice: the power it needs cannot be carried, and the power that can be carried does not hold the vehicle.

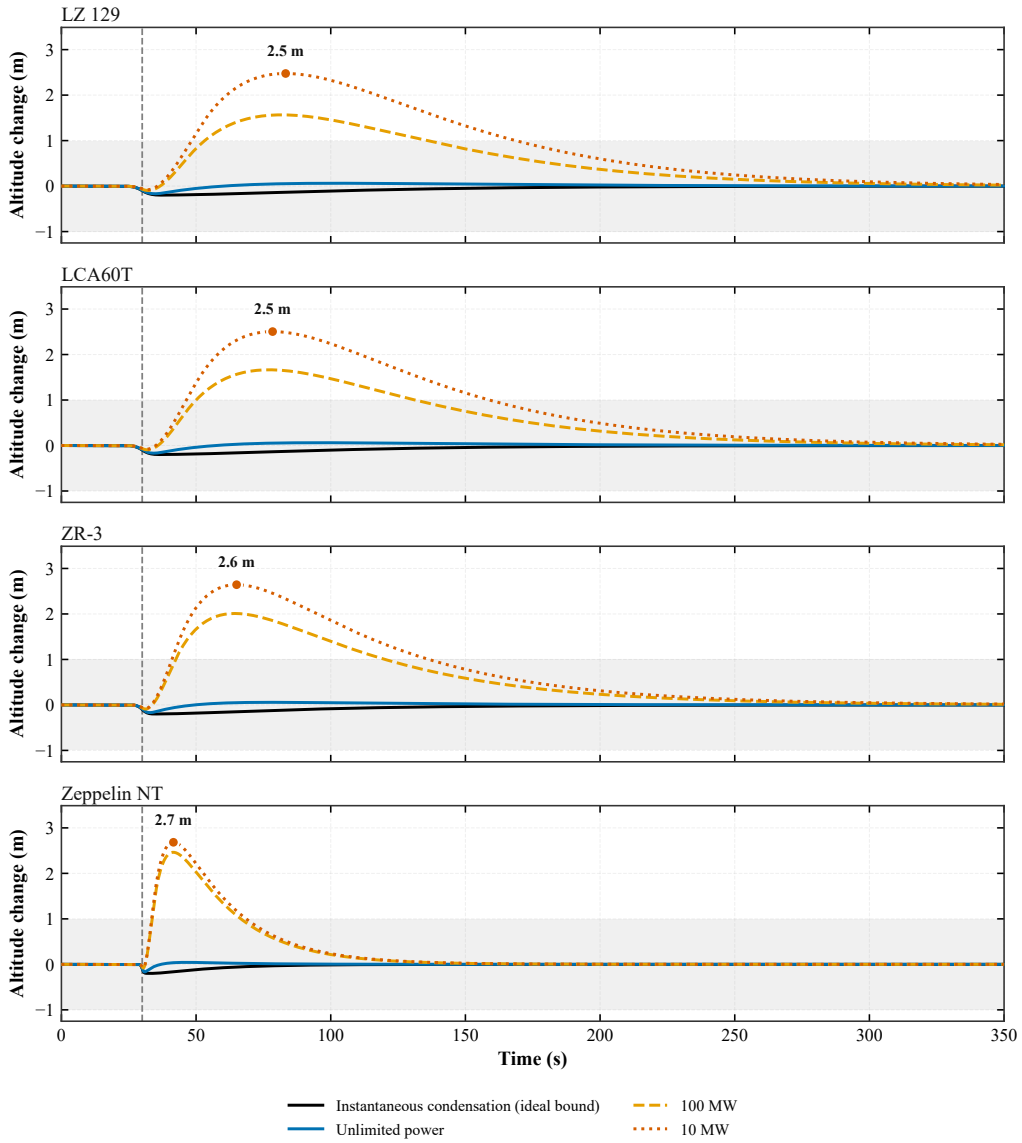


Figure 3: Altitude change of the four airships for a one-tonne release with the response of the heat-rejection unit, at 1 atm. Gray band: 1-m precision threshold; dashed line: release; dot: largest rise with 10 MW.

5.2. With a hold-down device

Restraining an airship is established practice: masts and handling lines hold it on the ground [2], the Zeppelin NT is certified to fly statically light on

its vectored propellers [23], and the LCA60T is designed to load and unload in hover [3], exchanging the payload against water ballast [40]. What this study adds is the coupling of such a device with the cycle. A hold-down device (HD) restrains the vehicle while the burn consumes the lifting gas down to the new equilibrium, at the pace of an installed heat rejection P_{cap}. Unless stated otherwise, the exhaust is at 14 atm in this section and in Sections 5.3 and 8; Figure 5 and Table 6 also give the case at 1 atm, without the exhaust compressor. At the release, the burn compensates by itself only the part of the release whose clean burn needs that power, $m_e = m_{\mathrm{load}}\,(P_{\mathrm{cap}}/P_{\mathrm{rej}})^{2/3}$: 80 kg of the tonne on the LCA60T with 10 MW. The device takes the rest, $(m_{\mathrm{load}} - m_e)\,g = 9.03$ kN, and then less and less: with 10 MW the burn removes 1.6 kg of buoyancy per second, so the force falls as a ramp until the vehicle reaches its new equilibrium after

$$t_{\mathrm{rec}} = \frac{m_{\mathrm{load}}\,q(0,\phi)}{L(\phi)\,P_{\mathrm{cap}}}, \tag{8}$$

10.6 min for the tonne with 10 MW, on any airship. The heat with no work exported, $q(0,\phi)$, is an upper bound of what the unit rejects when the engine exports work, so these recovery times are on the safe side. Without the exhaust compressor, at 1 atm, the same power recovers the tonne in 11.0 min, and more hydrogen leaves with the vapor that does not condense (Table 6), but the cycle saves the compressor (Table 5). At this burn rate the engine weighs 1.9–8.9 t and, after driving the fans and the exhaust compressor, still delivers power to the plant (Table 6).

The device can take two forms, which give the airship different capabilities. Propellers, such as the vectored propellers of modern airships, keep it autonomous: their peak power for the tonne, 0.06–0.19 MW for disk areas of 20–100 m^2 (figure of merit 0.7) over the installed heat rejections of Supplementary Table S10, is less than the engine delivers, so no battery is needed, but they must respond fast. A device that waits for the release must deliver the weight of the released load before the vehicle rises $h_{\max}$; with the vehicle accelerating at $\ddot{h} = m_{\mathrm{load}}\,g/(M_I - m_{\mathrm{load}})$, this bounds its response time:

$$\tau_{\mathrm{h}} \le \sqrt{\frac{2\,h_{\max}\,(M_I - m_{\mathrm{load}})}{m_{\mathrm{load}}\,g}}, \tag{9}$$

9.1 s for a tonne on the LCA60T and 1.9 s on the Zeppelin NT. As with the burn, anticipating relaxes the actuator: ramping its force at $\dot{F}_{\mathrm{h}} = L\,g\,\dot{m}^*_{H_2}$

before the release, the device on the Zeppelin NT tolerates a first-order lag of 2.3 s instead of 1.2 s (Supplementary Table S2); such a lag reaches the full force only after several time constants, so its time constant must stay below the bound of Equation (9). A tether, at bases or sites with suitable facilities, takes the same force with no power. Lowering the load by winch, so that the ground takes its weight gradually, has the same effect, with the lowering time as t_{rec} (Supplementary Table S1).

With the device, the release and its recovery time become a design choice. The installed heat rejection is $P_{\mathrm{cap}} = m_{\mathrm{load}}\, Q/t_{\mathrm{rec}}$, with $Q = 6.34$ GJ the heat per tonne released; the boxes that scale with it weigh $a = 2.1$ t per MW, the tanks and the regeneration stack $b' = 0.91$ t per tonne released, the water tank $b_{\mathrm{w}} = 1.3$ kg per tonne released, and the hydrogen compressor m_{c} comes in catalog units of 0.55 t, each of which compresses what 1.6 t of release regenerate in a day at 1 atm (Table 5). The payload C must hold them and the release itself, so the largest release that fits is

$$m_{\mathrm{max}} = \frac{C - m_{\mathrm{c}}}{1 + b' + b_{\mathrm{w}} + a\, Q/t_{\mathrm{rec}}}. \tag{10}$$

Since m_{c} adds one unit for every 1.6 t released, Equation (10) is solved by iteration for larger releases. The heat rejection limits short recoveries, and the boxes that grow with the release limit long ones. Since a, b', b_{w} and Q do not depend on the airship, the curves of Figure 4, divided by the payload, nearly collapse onto one on the three larger airships (Supplementary Figure S6, single release), each of which enters only through the fraction of its payload that a tonne represents; on the Zeppelin NT a single compressor unit takes over half the payload, and its curve falls below. On the three larger airships the fraction reaches half its ceiling after 1.5–1.7 h and never exceeds 41–44 %, however long the recovery. Large single releases are therefore out of reach: the LCA60T can release at most 27 t at once, so a thirty-tonne container does not fit whatever the recovery, and the Zeppelin NT at most 0.24 t; divisible cargo fares much better (Section 5.3). Without the exhaust compressor, at 1 atm, the boxes that scale with the heat rejection are lighter and the heat per tonne is larger; within the same payload the same release then recovers 15 % sooner on every airship, and the ceiling does not change.

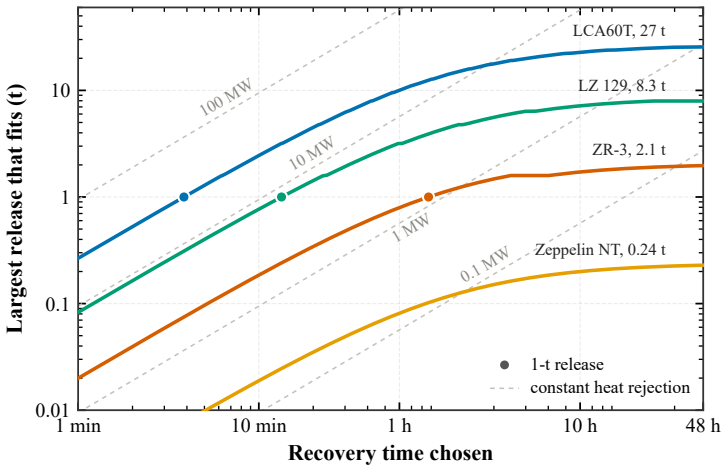


Figure 4: Largest release that fits the payload of each airship versus the recovery time chosen, with a hold-down device and the exhaust at 14 atm (Equation (10)). Dots: one tonne; dashed lines: constant heat rejection. The steps come from the hydrogen compressor, whose catalog units each serve 1.6 t of release: past each multiple, the release grows again only when the recovery is long enough for one more unit to fit.

Figure 5 gives the margins on each airship where each curve of Figure 4 reaches one tonne (HDM bars): the boxes and the tonne fill the payload, and each bar is labeled with the largest heat rejection that fits and the recovery time it gives. The LCA60T recovers the tonne fastest, in 3.9 min, and the recovery lengthens as the payload shrinks. On the Zeppelin NT the fixed boxes and the tonne already exceed the payload: a release of 0.2 t, the same share of its payload as a tonne on the ZR-3, takes 10 h, because one compressor unit takes over half the payload. With the reference 10 MW, the boxes and the tonne weigh 23 t on any airship, within the payload of the LCA60T alone. These are optimistic bounds: with the whole power-plant condenser instead of its bundles, the LCA60T would fit 7.2 MW and the tonne would take 15 min.

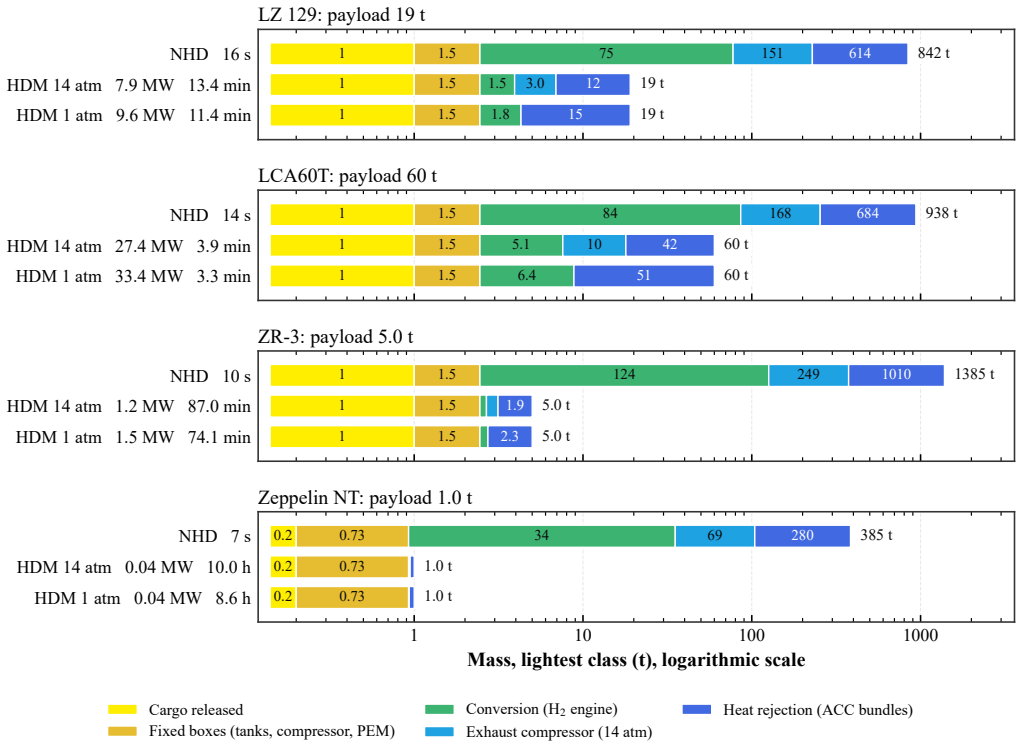


Figure 5: Mass carried for a release of 1 t (0.2 t on the Zeppelin NT) on the four airships: no hold-down device (NHD) and hold-down device with the largest heat rejection that fits the payload (HDM), with the exhaust at 14 atm or, without its compressor, at 1 atm. Logarithmic scale; masses in tonnes.

The device has its own limits. This study takes the airships by their hulls and payloads, not by their installed systems, and sizes the device as a black box; their actual propellers may fall short of it. The Zeppelin NT, for instance, flies with vectored propellers certified for a static lightness of at most 200 kg [23], below the 0.24 t it could release at once. Releases of thirty tonnes, beyond the ceiling of every airship, would also need 14–33 MW of propellers, over a hundred times the power that a tonne needs, which leaves them to moored airships where a tether serves as the device. With the device, then, the mass limit of real-time condensation becomes a time limit: the release that fits is set by the recovery time chosen, up to a ceiling set by the boxes that grow with the release.

5.3. Delivering in releases

A single release is the most demanding use of the device: the boxes fixed per tonne grow with the whole release, and the heat rejection must recover all of it. Divisible cargo can instead be delivered in consecutive releases, the vehicle recovering after each one with the same equipment. With releases of m_{load}, each recovered in t_{rec}, the installed heat rejection is $P_{\mathrm{cap}} = m_{\mathrm{load}}\,Q/t_{\mathrm{rec}}$, and the payload holds the equipment and the business cargo N delivered over all the releases. The water tank is the only box that grows with N rather than with the release, because the water of every release stays on board as ballast until it is regenerated; at 0.33–0.68 % of the water it holds, it is also the lightest:

$$N\,(1+b_{\mathrm{w}}) = C - m_{\mathrm{c}} - m_{\mathrm{load}}\left(b' + \frac{a\,Q}{t_{\mathrm{rec}}}\right), \qquad N \geq m_{\mathrm{load}}, \tag{11}$$

where m_{c} is now the compressor that one release needs, a single unit for releases of up to 1.6 t, which does not shrink with smaller releases. The bound $N = m_{\mathrm{load}}$ is the single release of Equation (10): each release size has a shortest recovery, and longer recoveries leave more of the payload for business cargo. In releases of one tonne the LCA60T delivers 36 t of business cargo with ten minutes per release and 51 t with thirty, against the 27 t of a single release, and the equipment for one-tonne releases of ten minutes weighs 24 t (20 t at 1 atm), which only the LCA60T can carry (Supplementary Figures S5 and S6 give every airship and release size against the recovery time of each release).

For a delivery, however, what counts is the time of the whole sequence, $T = n\,t_{\mathrm{rec}}$ for $n = N/m_{\mathrm{load}}$ consecutive releases. Written in it, Equation (11) takes the same form on every airship,

$$N = C'\,\frac{T}{(1+b_{\mathrm{w}})\,T + a\,Q}, \qquad C' = C - m_{\mathrm{c}} - b'\,m_{\mathrm{load}}, \tag{12}$$

so an airship delivers half of $C'/(1+b_{\mathrm{w}})$ in $a\,Q/(1+b_{\mathrm{w}}) = 3.7$ h of consecutive releases (3.1 h at 1 atm), and longer deliveries add less and less (Figure 6). The release size enters only through C': smaller releases leave more of the payload free, up to $C - m_{\mathrm{c}}$, while the single release bounds the family from below. What sets the cargo is therefore the pace of the delivery, $m_{\mathrm{load}}/t_{\mathrm{rec}}$, rather than the size of each release: one-tonne releases every ten minutes deliver 36 t in 6 h on the LCA60T (38 t at 1 atm), and half-tonne releases

every five minutes practically the same. The smaller airships follow the same curves scaled by their payloads, except that the catalog compressor takes a larger share of them, more than half on the Zeppelin NT.

The hold-down device sets which releases are practical. Propellers must hold the released load at once, less the little that the burn absorbs by itself, with a peak power that grows nearly as the released load to the power 1.5, whereas the engine that drives them grows with the installed heat rejection and is smaller the slower the recovery. On the LCA60T, with the exhaust at 14 atm, the engine powers propellers for one-tonne releases recovered in up to 29 min, or 6.7 h with the largest disks and an efficient engine, and for two-tonne releases in up to 20 min; at 1 atm the margin is wider (Section 6). Ten-tonne releases need the larger disks and the efficient engine, and twenty-tonne releases a tether or power beyond the cycle's engine, from the airship's own plant or a battery (Supplementary Section S8). On the small airships, the engine powers the propellers for their small releases with a wide margin. Delivered in releases of about a tonne, the cycle with a hold-down device and existing equipment thus carries most of the payload of the LCA60T as business cargo, one release every ten to thirty minutes, and half of that of the LZ 129 with thirty minutes per release (Supplementary Section S8). These figures take the condenser bundles, the optimistic bound of Table 5. With the whole power-plant system, $a\,Q/(1 + b_{\mathrm{w}})$ grows to 14.0 h: the LCA60T then delivers 18 t in 6 h, or 30 t in one-tonne releases of thirty minutes, and the LZ 129 needs an hour per release to deliver 3.6 t.

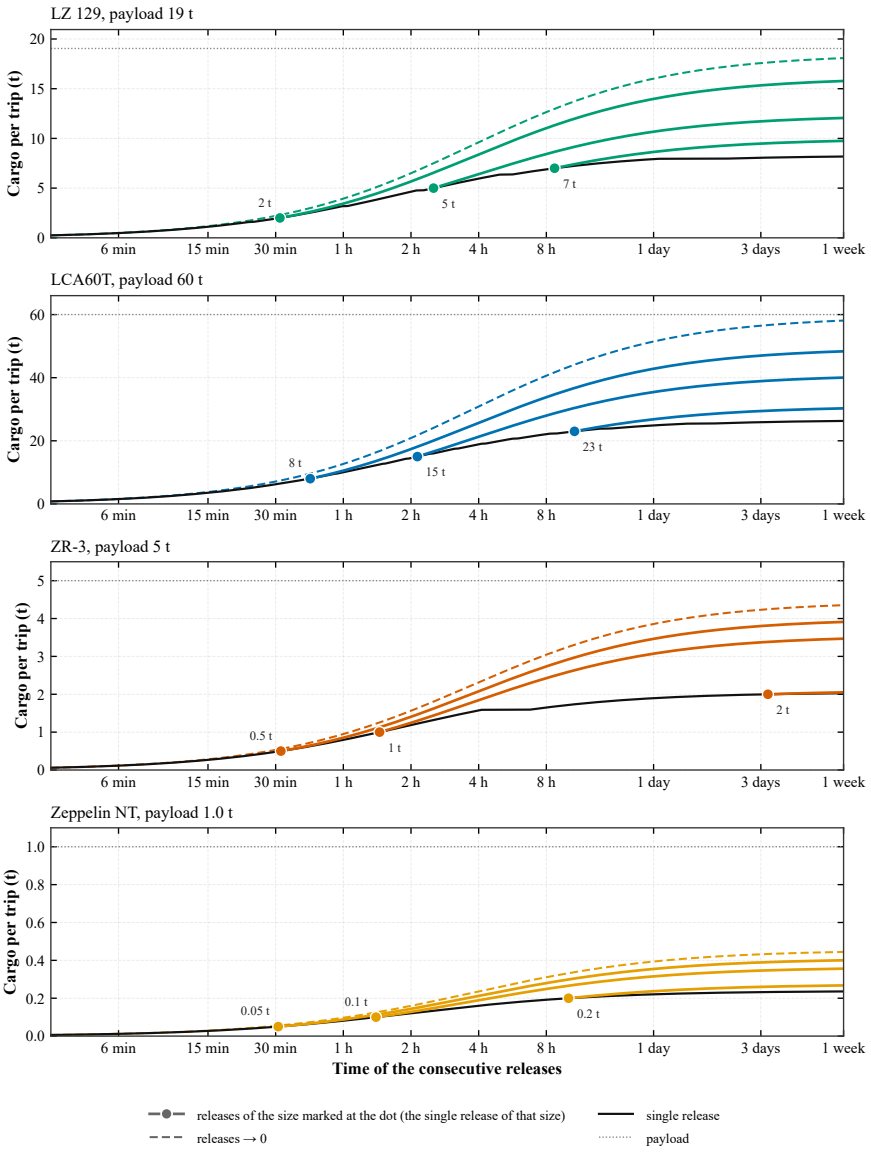


Figure 6: Business cargo per trip versus the time of the consecutive releases, one airship per row (Equation (12)), with the exhaust at 14 atm. Solid lines: releases of the size marked at the dot, where each starts as a single release; dashed, in the color of the airship: releases of vanishing size; black: single release; dotted: payload.

6. Energy and regeneration

After a release, electrolysis regenerates hydrogen from the retained water, on board or where electricity is at hand, while high-pressure tanks supply make-up hydrogen for the uncondensed fraction. As a representative class for the storage black box, the cycle uses Type IV vessels, scaled to 350 bar from the 700-bar DOE system [32, 41]; this study takes the composite mass proportional to pressure times volume. Their gravimetric capacity, the usable hydrogen over the mass of the complete storage system, is 5.4 %.

The release itself draws no energy from outside: it produces it. Burning the hydrogen for one tonne releases 1588 kWh at 1 atm, and the engine turns part of it into work that drives the fans and, as hold-down device, the propellers (at 14 atm also the exhaust compressor). Figure 7 shows this budget as power against the recovery time. The engine's work, the same energy spread over a longer time, falls as $1/t_{\mathrm{rec}}$, whereas the propellers must hold what the burn does not absorb at the release, whatever the pace. Propellers therefore suit short recoveries: the engine powers them for up to 1.1 h with the pessimistic engine, fans and disks, and 8.0 h with the optimistic ones; beyond that, a battery or the airship's own plant must supply them, while a tether holds the vehicle without energy at any pace. With 10 MW, part of that work is left over for the plant at either pressure (Supplementary Section S9).

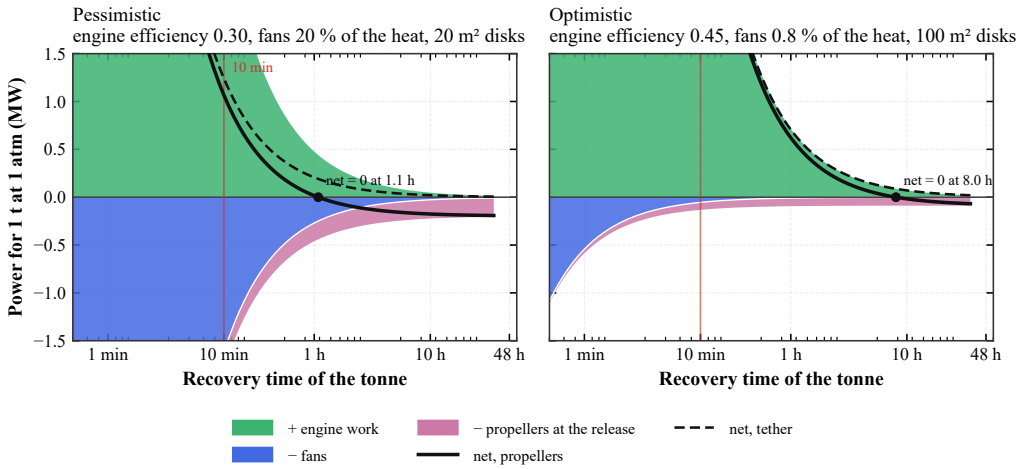


Figure 7: Power for a one-tonne release at 1 atm versus its recovery time, with the pessimistic (left) and optimistic (right) engine efficiency, fan power and propeller disk area. Above zero, the work of the engine averaged over the recovery; below, the fans and the propellers at the release. Solid: net power left for the plant with propellers (dot: where it reaches zero); dashed: with a tether. Red line: releases recovered in ten minutes.

Regeneration is where the energy goes. Splitting the retained water takes 2147 kWh of electricity per tonne released at 1 atm [42], and compressing the hydrogen into the tanks adds a few percent [43, 44]. The work left over from the release pays back only 9–31 % of this, so 1529–2079 kWh per tonne must come from outside, and 1869–2418 kWh at 14 atm, which retains more water and so regenerates more hydrogen. All of it must be supplied by ground infrastructure or onboard solar arrays.

Regenerating on board also has a mass. The stack, the compressor and the hydrogen and water tanks already travel in the payload budgets of Section 5, sized there to regenerate a release in a day with electricity from outside; in flight, what limits regeneration is that electricity. As a module powered only by the sun, the regeneration unit is rated by the hydrogen it returns per hour, at about 54.9–56.0 kWh per kilogram, with the classes of Table 5: 24–106 kg of proton-exchange-membrane (PEM) stack per kg/h [34] and one catalog compressor of 545 kg for every 2.69 kg/h [33]. The tanks waiting for the hydrogen add 17.5 kg per kilogram stored, and 18.5 at 700 bar, so a higher pressure saves volume, not mass. The hydrogen itself adds nothing, since it comes from the water already carried as ballast, and the oxygen vented while regenerating lightens the vehicle.

To evaluate regeneration in flight, this study assesses photovoltaic arrays covering a fraction 0.45 of the hull surface, on its upper half [45], with the irradiance measured in Lima, Peru [46] and transposed to the curved hull [47, 48]: 3.4 h of full sun per day on average, 2.0 in winter and 4.6 in summer. Two representative classes of flexible panels are evaluated as the source of that electricity, each sized for the peak power of the unit: bare amorphous-silicon films (8 %, 0.12 kg/m^2 [49], an optimistic value without encapsulation) and commercial copper indium gallium selenide (CIGS) modules (14.6 %, 2.4 kg/m^2 [50]). Figure 8 gives the mass of such a module against the hydrogen it recovers and the hours of sun it takes, for either class and any airship.

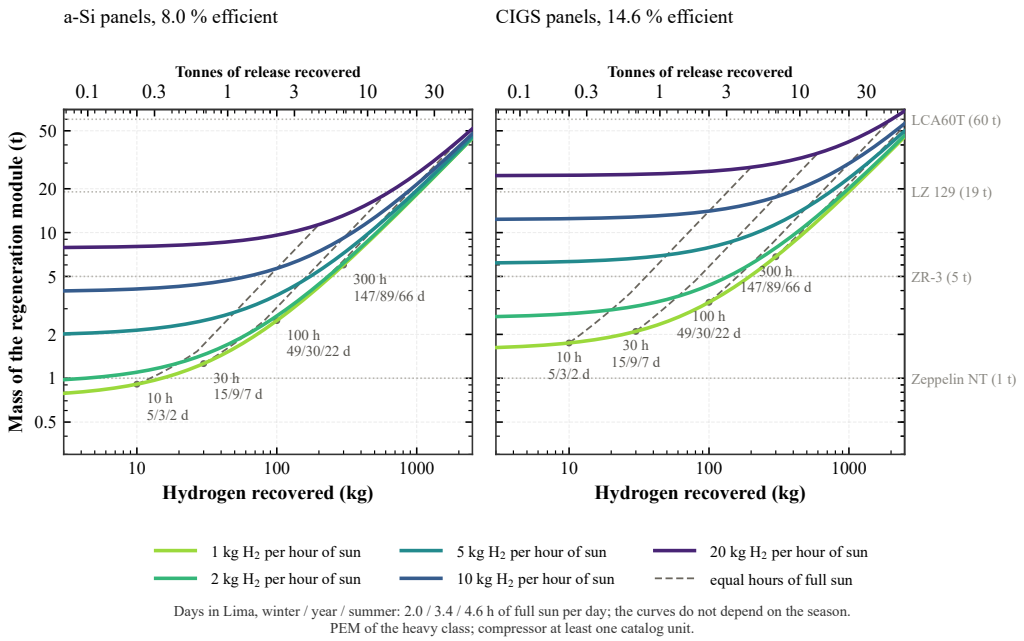


Figure 8: Mass of a regeneration module powered only by hull panels (panels, PEM stack, hydrogen compressor and tanks) versus the hydrogen it recovers (top axis: tonnes of release it returns), for the two representative classes of panels: left, amorphous-silicon films; right, CIGS modules. Solid lines: capacity in kilograms of hydrogen per hour of full sun; dashed: equal hours of full sun, with the corresponding days in Lima in winter, on average and in summer. Payloads of the four airships on the right, for reference.

Divided by the tonnes it returns, the module gives what regeneration in

flight costs in payload when it is allowed T_r days of sun:

$$\frac{m_\mathrm{mod}}{m_\mathrm{rec}} = k_\mathrm{t}\, m_\mathrm{r} + \frac{a_\mathrm{r}}{T_\mathrm{r}}, \tag{13}$$

where m_mod is the mass of the module, m_rec the lift it recovers, $m_\mathrm{r} = 44.6$ kg the hydrogen regenerated per tonne (at 14 atm) and k_t the tank mass per kilogram stored. The first term, the tanks, is a floor of 0.78 t of payload per tonne recovered whatever the time; the second, the regeneration unit and its panels, with a_r their mass per tonne recovered and per day of sun, shrinks as the time allowed grows. Within a week, each tonne recovered costs more payload than it returns with either class of panels (Supplementary Section S9); it costs less only after 19–24 days of average sun with amorphous silicon, and after 1.9–2.6 days with electricity available all day, as at a site with a grid connection; that is the time the module takes to pay back its own mass. Regenerating in flight is therefore possible but small: the hours of sun that the flight leaves free set both what the module recovers and its mass (Figure 8), and the rest of the hydrogen is regenerated where electricity is at hand. As black boxes, the panels count only for the electricity they deliver and their mass; Section 9 discusses their effect on the heat balance of the hull.

Regenerating in flight also frees lift: venting the oxygen of the water lightens the vehicle by 354 kg per tonne recovered at 14 atm. Whether that lift has to be removed is a design and logistics choice, like the hours of sun: before loading it is the lift wanted, and keeping the oxygen on board avoids it, at the cost of vessels of its own. Removing it by compressing lifting gas into the tanks takes 24.6 kg of hydrogen per tonne on top of the regenerated hydrogen, so the tanks of Table 5 suffice for 71–75 % of the hydrogen of a release. Tanks sized to hold it all would raise the floor of Equation (13) to 1.21 t of payload per tonne recovered, more than the lift they help recover.

7. Comparison with other methods

Table 6 compares the cycle against alternative buoyancy control methods. Venting loses the gas; water ballast requires ground supply networks; sliding ballast provides no net heaviness; and conventional vacuum chambers are too heavy [6]. Compressing the lifting gas into onboard tanks, the principle of COSH [7], is a direct alternative; for hydrogen, Table 6 gives only the balance of compressing the gas instead of burning it, not a system.

During a release, all methods must operate over the same duration (14 s for a one-tonne release on the LCA60T). This is because the clean maneuver must remove lift at a pace set by the released load, the inertia of the airship and the sink allowed before the release, regardless of which gas is being processed (since $m_{\text{load}}/(L\,\dot{m}^*_{H_2})$ does not depend on L, Equation (7)). The methods therefore differ only in the gas they process, and their power scales with their energy per tonne. With a hold-down device, every method is compared over the 10.6 min that the cycle requires at an installed heat rejection of 10 MW (which also does not depend on the airship, Equation (8)). Another recovery time changes only the power rows, which scale as the energy per tonne over the time chosen; the rows per tonne hold for any recovery time and airship.

Helium compression requires 14 times less energy than the cycle's heat rejection at 14 atm, and 15 times less at 1 atm. It also rejects that heat at a high temperature, where it is easier to dissipate, but it draws power from the plant; the cycle rejects its heat in the demanding low-temperature condensation band but, with an engine, supplies power to the plant. A hold-down device lowers the power requirement for all methods equally. To restore lift, compression returns all its gas at once. The cycle returns at once only the burned hydrogen, 65 % of the lift at 14 atm and 68 % at 1 atm; the rest recovers slowly through electrolysis, in about a day per tonne with electricity from outside (Table 6).

The regenerative cycle trades real-time heat rejection, electrolysis, and slow lift recovery for ambient-pressure ballast storage. It requires no pressure vessels to store the removed lift (for COSH with helium, the composite alone would weigh 39 t for the full LCA60T payload). The difference grows over a delivery: compression must store the lift of every tonne delivered, so its tanks grow with the whole cargo, whereas the cycle keeps that lift as water and reuses its equipment in every release. To deliver 36 t on the LCA60T in one-tonne releases, compressing hydrogen would need at least 26 t of tank composite, against 24 t for all the equipment of the cycle. The cycle's own tanks return only a fraction of the lift at once, and the rest comes back as the water is regenerated, which needs electricity but no hydrogen supply on the ground: slowly in flight with panels, faster with a grid connection (Section 6). Against direct venting, it avoids 91 % of the hydrogen loss at 1 atm and 99 % at 14 atm. Counting the work left by the release, its regeneration needs 47–61 % less external energy than ground resupply at 1 atm, and 39–53 % less at 14 atm.

Table 6 is organized in two stages, the release and the restoration of the lift, and the cycle appears twice: retaining 0.99 of the water, with the exhaust compressed to 14 atm, and retaining 0.85, the most that condenses at 1 atm and 40 °C (Table 4). Only the power "at once" depends on the airship; every other row is per tonne and holds for any of the airships. The caption gives how each row is computed.

Table 6: Alternative buoyancy control routes per tonne released. 0: applies and is zero; –: does not apply or is not evaluated. Power at once: the heat rejected over the release (for compression, the minimum isothermal work, which also leaves as heat). Net power: an engine efficiency of 0.30–0.45, fans taking from 0.008–0.041 of the rejected heat (sized for the pressure drop) to 0.20 (aircraft radiators) and, at 14 atm, the exhaust compressor driven by the engine; compression draws its work, with a compressor not sized in this study. Tank composite: a lower bound, scaled from the Type IV system of [32] with pressure times volume, so it barely depends on the storage pressure chosen (0.61–0.68 t per tonne for helium over 5–30 MPa); hydrogen is stored at the pressure of the cycle's tanks. Energy to restore the lift: electrolysis and compression of the regenerated hydrogen or, for venting, electrolysis of the resupplied gas. Real-gas properties for the compression work and the tank volumes: helium from [51], hydrogen from [52].

	Hydrogen				**Helium**	
Per tonne released	**burned** ($\phi = 0.99$)	**burned, 1 atm** ($\phi = 0.85$)	**compressed** (35 MPa)	**vented**	**compressed** (COSH, 20 MPa)	**vented**
Release						
Heaviness per kg of gas, L	22.22	20.98	14.37	13.37	7.24	6.24
Gas processed	45.0 kg	47.7 kg	69.6 kg	74.8 kg	138.2 kg	160.3 kg
Heat to reject	6.34 GJ	6.57 GJ	0.50 GJ	0	0.45 GJ	0
Power, at once (14 s)	446 MW	464 MW	35 MW	0	31 MW	0
Net power to the plant, with a hold-down device (10.6 min)	+0.5–3.3 MW	+1.2–4.0 MW	−0.8 MW	0	−0.7 MW	0
Lift removed, stored as	water 0.40 m^3	water 0.36 m^3	gas 2.9 m^3	–	gas 4.5 m^3	–
Restoring the lift						
Gas returned from onboard tanks	45.0 kg	47.7 kg	69.6 kg	–	138.2 kg	–
Lift restored at once	65 %	68 %	100 %	0	100 %	0
Composite of those tanks, at least	0.47 t	0.50 t	0.73 t	–	0.65 t	–
Rest of the lift restored by	electrolysis, O_2 to the air	electrolysis, O_2 to the air	–	ground resupply	–	ground resupply
Time to restore the rest	26 h	24 h	–	–	–	–
Energy in this stage	2.4–2.5 MWh	2.2–2.3 MWh	0	4.0 MWh	0	–
Gas lost	0.5 kg	7.1 kg	0	74.8 kg	0	160.3 kg

8. Verification, validation, and sensitivities

Code verification and component validation. The model is verified against closed-form solutions, and each phenomenon ranked high in Supplementary Table S2 is checked against the best evidence available for it. It reproduces them within 0.003 % in altitude, with numerical uncertainties below 0.002 %.

Of the seven phenomena ranked high, three rest on data: the condensation bound holds against measurements, and data bound the vertical inertia and the masses of the black boxes. Of the other four, the thermochemistry is checked against reference data, the outside-air temperature is covered by sensitivity, the hold-down device rests on a closed form and simulation, and the heat rejection is the output being specified. The condensation bound was compared with the water recovered from an open-cathode hydrogen fuel cell [53]. Its analytical balance matches their theoretical curves within 0.12 percentage points (Supplementary Section S4.3), and the cell collected 68–85 % of the theoretical value, consistent with the model being an upper bound for real condensers. No measurement exists for the integrated release, since no airship of this kind has been built; the warnings of Section 10 state what that leaves open.

Bounds, sensitivities, and credibility. In the sensitivity analyses, vertical drag changes the peak excursion by only about 0.5 %. The largest epistemic uncertainties are practical water retention and vertical inertia, which shift the installed burn rate by 12–20 % and by up to 14 %, respectively. None of these variations brings the heat rejection for a one-tonne release below several hundred megawatts. With a hold-down device, a growth margin of 10 % on the mass of all the equipment, nominal here (Section 3.5), costs 2.4 t of the 36 t that the LCA60T delivers in one-tonne releases of ten minutes, and the same margin on the heat to reject 2.2 t: the cargo shrinks, but the delivery holds. Supplementary Sections S4–S6 give the details of the code verification, the parameter bounds and the authors' credibility assessment with NASA-STD-7009B, which documents the boundaries of the model's validity.

9. Discussion

Inertia. The vertical inertia is the least certain block. Constrained and free-flight tests of an 18-m^3 blimp [54] bracket the potential-flow estimate of its heave mass, vehicle plus added mass (Supplementary Section S4). That spread changes the dynamic requirements by −9 % to +14 %, leaves the static ones unchanged, and keeps the altitude loop overdamped.

Climate and combustion. Hot climates reduce the water that can be retained (Table 4), and real combustors with excess air dilute the vapor further. Both push toward condensing at a higher pressure. Operating higher does not change this much: with the standard lapse rate from the design hot

day, at 1 km the air is at 23.5 °C, the water that can be retained rises from 85 to 89 % and the hydrogen per tonne falls by 1.4 %, while the same airship loses 9 % of its lift. Real profiles, such as coastal inversions, must be taken from local soundings.

Condensing at pressure. A higher exhaust pressure moves the condensation band to higher temperatures, farther above the cooling air. At 5 atm the band runs from 116 to 74 °C, and the exhaust still retains as much water as at 1 atm. The compression takes at least 5.4 MJ per kilogram of hydrogen, and whether such an intermediate pressure pays depends on the mass of the exhaust compressor, which this study scales only for 14 atm (Table 5).

Isothermal gas. The lifting gas is taken as isothermal during the maneuver. Its thermal time constant in a hull of the size of the LCA60T is about 27 min, and 31 min with the airship loaded at the start of a delivery (natural convection [55] with the properties of [51]), so a change of 5–20 K in solar superheat during the sized burn shifts the buoyancy by only 16–65 kg, a few centimeters of travel, well inside the 1-m threshold. Over the recovery of a one-tonne release with a hold-down device, a change of 5–10 K shifts it by 0.6–1.2 t, of the order of the release itself. A whole delivery lasts hours, long enough for the gas to settle at the new temperature, so the same change shifts the buoyancy of the loaded airship by 3.0–6.0 t, several releases' worth (Supplementary Table S2). Hold-down recoveries and deliveries in consecutive releases therefore assume steady solar conditions, or must treat thermal drift as a further, slow disturbance for the same buoyancy control.

A buoyancy attenuator. With a hold-down device the cycle converts lift into water ballast at the pace that its heat rejection allows, while the device holds the vehicle. Moored, the tether is the device: it responds at once and draws no energy, so the cycle can trade heat rejection for time without limit. This opens uses beyond the release in flight. The first is exchanging a payload without ballast water or hydrogen supplied on the ground, only electricity to regenerate afterwards; the manufacturer of the LCA60T has patented a payload exchange against water ballast pumped from a tank on the ground at the delivery site [40]. The airship then needs a mast to moor, and the payload taken on must wait for the electrolysis to return the rest of the lift, or the tanks must hold the hydrogen to return all of it (Section 7). The cycle can also trim the lift slowly, against solar superheat or before landing, instead of venting gas, and deliver a payload in consecutive releases. The cycle remains an accumulator: the hydrogen burned must be regenerated (Section 6), and releases in free flight beyond the authority of the device

keep the real-time requirement of Section 5.

Regeneration in flight. Section 6 evaluates regeneration in flight with the sun of one site, Lima, but draws Figure 8 in hours of full sun, so it holds at any site: the site changes only the days that those hours take, and neither the floor of the tanks nor the external electricity per tonne depends on it. Panels on the hull would also change its heat balance, and with it the thermal drift above, by an amount that depends on their installation and is not assessed here.

Safety. Hydrogen is not admitted as a lifting gas by current airship airworthiness codes such as the Canadian one [56], so certifying a hydrogen airship is a prerequisite of the cycle and outside this study. Burning hydrogen on board places a flame, up to 2256 °C adiabatic without dissociation, next to the gas cells, so the conversion unit, its exhaust ducts and the heat-rejection unit must be isolated from the cells and from the hydrogen they draw; this belongs to the design of the black boxes and is not assessed here. So do the oxygen released by the electrolysis, which must be vented away from the cells, and leaks at the regulator that feeds them; together they call for a hazard analysis of the installation, such as the safety assessment process of SAE ARP4761A [57], which is future work. Venting, the alternative, carries its own risk: in hydrogen airships, venting gas to lose lift on the descent, together with static discharge at ground contact, made the approach to the ground especially hazardous [58]; at release, the cycle replaces that venting with a controlled burn.

10. Conclusions

1. With instantaneous condensation, the installed burn rate follows in closed form, $\dot{m}^*_{H_2} \propto m_{\text{load}}^{3/2}/\sqrt{M_I}$: 3.17 kg/s of hydrogen for one tonne on the LCA60T, with a preconditioning fraction $\alpha^* = 0.5$ for every airship (Table 3). The sized burn ends at rest at the new equilibrium, and the added mass, which nearly doubles the vertical inertia, must be part of the dynamics.
2. Retaining the water, not burning the hydrogen, governs the cycle: retaining 0.99 of it means rejecting 141 MJ per kilogram of hydrogen, 446 MW in real time for a one-tonne release on the LCA60T. The condensation band just above the cooling air holds only 14 % of that heat at 1 atm, but it sets the size of the heat-rejection unit, which must be running before the release.

3. The energy closes but the mass does not. A conversion efficiency of 0.15–0.19 powers the cooling at 1 atm, but, with the black boxes of this configuration, the heat-rejection unit must then reject at least 8–11 kW/kg, and the classes that condense at the temperature of the cycle fall 3–64 times short; at 14 atm, the exhaust compressor alone outweighs the payload. Real-time condensation is therefore the development gap for buoyancy control alone.
4. A hold-down device, such as the vectored propellers that the airship already carries or a tether when moored, turns the real-time requirement into a choice of recovery time: with 10 MW and the exhaust at 14 atm, the tonne recovers in 10.6 min on any airship, with 15 t of condenser bundles, an optimistic bound. Propellers need little power for a tonne, but container-scale releases need 14–33 MW, so only a moored airship can hold them.
5. Delivering divisible cargo in consecutive releases multiplies what fits, since the same equipment serves every release. The pace of the releases, not their size, sets the cargo (Equation (12)): on every airship, half of the payload left free by the equipment arrives in 3.7 h of releases at 14 atm (3.1 h at 1 atm), and the LCA60T carries 36 t in 6 h of one-tonne releases, against 27 t released at once. Propellers powered by the engine hold releases of one to two tonnes, and of ten with the largest disks and an efficient engine; larger ones need a tether or power beyond the cycle's engine.
6. Against venting, the cycle keeps 91 % of the hydrogen at 1 atm and 99 % at 14 atm, but its regeneration needs 1529–2079 kWh of external electricity per tonne at 1 atm; in flight it is possible but small, sized by the hours of sun that the flight leaves free. Against compressing the lifting gas, it needs about 14–15 times more heat rejection than helium and restores only part of the lift at once, but it stores the lift it removes as water at ambient pressure, without pressure vessels.

These requirements carry six warnings, since the model is verified but only two of its blocks are checked against measurements: the vertical dynamics of a full-size airship are not validated (−9 to +14 % on the dynamic requirements); the condensation bound and the black box of the heat-rejection unit favor the cycle, so the requirements are lower bounds and a real unit may retain less water; the installed rate carries no margin for timing, mass estimation or gusts; the equipment masses carry none for growth (2.4 t of the

36 t delivered for a margin of 10 %); the thermal drift of the lifting gas adds a slow disturbance to hold-down recoveries and deliveries (Section 9); and the propellers are ideal, with the response of the device not validated.

Real-time heat rejection at the temperature of the cycle is the gap between the ideal bound and a working cycle for buoyancy control alone. Its quantification—the heat per kilogram and per release, the rejection temperature, the power, the response time and the mass per kilowatt—is the specification for equipment development, experimental validation of the vertical dynamics and gradual payload lowering; with a hold-down device, the specification adds the force, the response and the energy of the device, which set the releases that can be held in flight.

CRediT authorship contribution statement

Antony Elthon Espinoza Leon: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Visualization. **Jorge Enrique Ortiz Porras:** Validation, Formal analysis, Supervision, Writing – review & editing. **Flor de María Laura Ortiz Pinedo:** Conceptualization, Investigation, Data curation, Writing – review & editing. **Rosiand Manrique Alva:** Investigation, Writing – review & editing, Validation. **Francisco Javier Wong Cabanillas:** Data curation, Supervision, Writing – review & editing. **Alcides Guillermo Joo Aguayo:** Software, Formal analysis, Writing – review & editing. **Gina Coral Tejada Estrada:** Methodology, Data curation, Resources, Writing – review & editing. **Ysabel Zevallos Parave:** Validation, Methodology, Writing – review & editing. **Fidel Tadeo Soria Cuellar:** Project administration, Resources, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with the development and testing of the analysis code, the com-

parison of model inputs against published reference data, and the drafting and English-language editing of the manuscript. The authors directed the analysis, checked the results and the text, and take full responsibility for the content of the published article.

Data availability

The code that generates every number, table and figure of this article, with its verification checks and the register of the sources used, will be made openly available in the GitHub repository https://github.com/Rodia23/regenerative-h2-cycle-airships-ast and archived in Zenodo, https://doi.org/10.5281/zenodo.23396619. Simulations were implemented in Python 3.14 using NumPy, SciPy and Matplotlib. The supplementary material, generated by the same pipeline, gives the credibility assessment of the model with NASA-STD-7009B: its assumptions, the phenomena ranking, the reference-data checks, the details of the validation, the sensitivities, the level of each factor and the warnings.

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Appendix A. Derivation of the ideal bound

Assumptions. The derivation takes the inertia constant within each phase, the air density constant and the drag negligible (the vehicle moves at centimeters per second), and releases the payload as a step change in load; the simulations of Section 4 keep the step change and relax the other three.

Preconditioning. At equilibrium the buoyancy $\rho_{\text{air}} g V_0$ balances the weight Mg. Burning hydrogen at a constant rate $|\dot{m}|$ with the water retained removes gas volume $|\dot{m}|\,t/\rho_{H_2}$ and adds mass $(r-1)|\dot{m}|\,t$, so the net force is $\Delta F(t) = -L\,g\,|\dot{m}|\,t$ with L from Equation (3) at $\phi = 1$. Dividing by the inertia M_I and integrating twice from rest,

$$|\Delta h(t)| = \frac{L\,g\,|\dot{m}|}{6\,M_I}\,t^3. \tag{A.1}$$

Limiting the sink to D_{lim} sets the anticipation time and the fraction of the payload that the hydrogen burned in it compensates,

$$t_{\mathrm{anticip}} = \left(\frac{6\, M_I\, D_{\mathrm{lim}}}{L\, g\, |\dot{m}|} \right)^{1/3}, \qquad \alpha = \frac{L\, |\dot{m}|\, t_{\mathrm{anticip}}}{m_{\mathrm{load}}}. \tag{A.2}$$

Solving for the rate that reaches a given α,

$$\dot{m}_{\mathrm{pre}}(\alpha) = K_{\mathrm{pre}}\, \alpha^{3/2}, \qquad K_{\mathrm{pre}} = \frac{m_{\mathrm{load}}^{3/2}\, \sqrt{g}}{\sqrt{6\, M_I\, L^2\, D_{\mathrm{lim}}}}. \tag{A.3}$$

After the release. The preconditioning burn delivers the vehicle to the release moving downward, with momentum $M_I\, |v_0| = L\, g\, \dot{m}\, t_{\mathrm{anticip}}^2/2$, that is $|v_0| = 3\, D_{\mathrm{lim}}/t_{\mathrm{anticip}}$. After release the inertia is $M_I - m_{\mathrm{load}}$ and the net upward force, $g\,[m_{\mathrm{load}} - L\, \dot{m}\, (t_{\mathrm{anticip}} + t)]$, falls linearly to zero when the hydrogen burned reaches the new equilibrium, m_{load}/L, a time t_r after release; over that time it supplies the upward impulse $L\, g\, \dot{m}\, t_r^2/2$.

Clean burn. The velocity is continuous at release, so the vehicle comes to rest exactly at the new equilibrium if $(M_I - m_{\mathrm{load}})\, |v_0| = L\, g\, \dot{m}\, t_r^2/2$, that is $t_r = s\, t_{\mathrm{anticip}}$ with $s = \sqrt{1 - m_{\mathrm{load}}/M_I}$. The hydrogen burned before release is then the fraction $\alpha^* = t_{\mathrm{anticip}}/(t_{\mathrm{anticip}} + t_r) = 1/(1+s)$ of the total, and $\dot{m}_{H_2}^* = \dot{m}_{\mathrm{pre}}(\alpha^*)$ gives Equation (7). The velocity rises monotonically from v_0 to zero after release, so the vehicle sinks a further $s\, D_{\mathrm{lim}}$ and never rises: with the upward force $L\, g\, \dot{m}\, (t_r - t)$ acting on $M_I - m_{\mathrm{load}}$ from $v = -|v_0|$, integrating twice over t_r gives a further sink $|v_0|\, t_r/3 = s\, D_{\mathrm{lim}}$. With $s = 1$, the same inertia before and after release, Equation (7) reduces to $\dot{m}_{H_2}^* = \sqrt{g\, m_{\mathrm{load}}^3/(48\, M_I\, L^2\, D_{\mathrm{lim}})} \propto m_{\mathrm{load}}^{3/2}/\sqrt{M_I}$, the scaling of Section 4; every simulation and table uses Equation (7) itself, which differs from this approximation by at most 0.2 % on the large airships and 2 % on the Zeppelin NT, whose payload is a larger share of its inertia (Table 2). The inertia of the air that replaces the burned hydrogen is neglected here, an effect of the same order as the correction s, so Equation (7) is exact only within the assumptions above; the simulations include it, and their peak depth differs from $(1+s)\, D_{\mathrm{lim}}$ by at most 2 mm, below the resolution of Table 3.

Appendix B. Altitude controller

The feedback law commands a burn or, when the vehicle sinks below its equilibrium, an injection from the storage tanks, proportional to the volume

error,

$$\dot{m}_{\mathrm{PD}} = K_m \left(V_{\mathrm{eq}} + K_p \left(h_0 - h \right) - K_d \, v - V_{\mathrm{gas}} \right), \qquad K_d = 2 \, \zeta \sqrt{\frac{K_p \, M_I}{\left(\rho_{\mathrm{air}} - \rho_{\mathrm{H_2}} \right) g}}, \tag{B.1}$$

where V_{eq} is the equilibrium gas volume for the current mass, which the controller is assumed to know exactly (the release instant, the payload released and the water retained; estimating it is left to future work), $K_p = 15 \ \mathrm{m^3/m}$ and $\zeta = 1.30$. A negative command burns hydrogen and a positive one returns it from the tanks, so the burn rate of the rest of the paper is $-\dot{m}_{\mathrm{PD}}$. The derivative gain uses the actual inertia M_I of Equation (2), so the damping ratio is constant throughout the maneuver. The gain uses the restoring force of the loop, $(\rho_{\mathrm{air}} - \rho_{\mathrm{H_2}}) \, g$ per unit of gas volume, since each cubic metre of gas in the cells lifts its buoyancy less its own weight, so ζ is the damping ratio of the loop. K_p sets a slow loop, with a natural period of several minutes, which leaves the fast part of the maneuver to the saturated burn, and the loop is overdamped so that the trim does not overshoot the reference altitude. The gain K_m is 60 $\mathrm{kg\,s^{-1}\,m^{-3}}$ until the gas mass reaches its new equilibrium and 0.5 $\mathrm{kg\,s^{-1}\,m^{-3}}$ afterwards. During the anticipation window the controller is replaced by the preconditioning burn at the installed rate, and the command is saturated at $\pm \dot{m}^*_{H_2}$, so hydrogen may be returned from the tanks at up to the installed rate (Table 1).

Appendix C. Heat to reject and minimum conversion efficiency

Heat per kilogram of hydrogen. With no work exported and $\phi = 0.99$, the terms of Equation (4) are

$$\begin{aligned} q_0 &= \mathrm{LHV} + \phi \, r \, L_v + \Delta h_{\mathrm{in}} - \Delta h_{\mathrm{out}} \\ &= 119.96 + 21.60 + 0.244 - 0.969 = 140.84 \ \mathrm{MJ/kg}, \end{aligned} \tag{C.1}$$

where Δh_{in} is the sensible heat that the reactants bring above 25 °C and Δh_{out} the sensible heat that the exhaust takes away at $T_{\mathrm{out}} = 40$ °C; together they are Δh_{sens}. Of this heat, 26.2 MJ/kg (19 %) is released between the dew point, 151 °C at 14 atm, and T_{out}. At 1 atm, with $\phi = \phi_{\max}$, $q_0 = 137.9$ MJ/kg, of which 19.8 MJ/kg (14 %) is released between the 73 °C dew point and T_{out} (Section 5.1).

Minimum conversion efficiency. The work $\eta\,\mathrm{LHV}$ must drive the fans, which take the fraction f of the heat rejected, and any compressor of the exhaust, whose work per kilogram of hydrogen is W_{comp}. The compression work also leaves as heat, so with work exported the heat to reject is $q_0 - \eta\,\mathrm{LHV} + W_{\mathrm{comp}}$, and the cycle powers its own cooling if

$$\eta\,\mathrm{LHV} \geq f\,(q_0 - \eta\,\mathrm{LHV} + W_{\mathrm{comp}}) + W_{\mathrm{comp}}, \tag{C.2}$$

which, solved for η, gives

$$\eta \geq \eta_{\min} = \frac{f\,(q_0 + W_{\mathrm{comp}}) + W_{\mathrm{comp}}}{(1+f)\,\mathrm{LHV}}. \tag{C.3}$$

Nomenclature

A	Drag reference area, the planform of the hull (m^2)
a	Mass of the boxes that scale with the installed heat rejection, Equation (10) (t/MW)
a_{r}	Regeneration unit and panels per tonne recovered and per day of sun, Equation (13) (t·day/t)
b'	Tanks and regeneration stack per tonne released, Equation (10) (t/t)
b_{w}	Water tank per tonne released or delivered, Equations (10) and (11) (t/t)
C	Payload of the airship (t)
C'	Payload left after the hydrogen compressor and the boxes of one release, Equation (12) (t)
C_D	Drag coefficient
D	Hull diameter (m)
D_{lim}	Maximum sink during preconditioning (m)
f	Power to move the cooling air per unit heat rejected
$\dot{F}_{\mathrm{h}}$	Rate of force of a hold-down device that anticipates the release, Section 5.2 (N/s)
FM	Figure of merit of a propeller in static thrust
g	Standard gravity ($\mathrm{m/s}^2$)
h, h_0	Altitude and reference altitude (m)
$\ddot{h}$	Vertical acceleration ($\mathrm{m/s}^2$)
$\ddot{h}_{\mathrm{s}}$	Acceleration that the release imposes, $m_{\mathrm{load}}\,g/M_I$ ($\mathrm{m/s}^2$)

$h_{\max}$	Largest displacement from h_0 allowed after release, upward or downward (m)
H_s	Density scale height of the atmosphere (km)
h_w	Heat to reject per kg of condensed water (J/kg)
Δh_{sens}	Sensible heat of reactants and exhaust per kg of hydrogen (J/kg)
Δh_{in}, Δh_{out}	Sensible heat that the reactants bring above 25 °C; that the exhaust takes away, Appendix C (J/kg)
ΔF	Net force of the burn before the release, Appendix A (N)
j	Jerk of the burn, $L\,g\,\dot{m}^*_{H_2}/M_I$ (m/s^3)
k	Transverse added-mass coefficient, Figure 2
k_{t}	Tank mass per kg of hydrogen stored
K_d	Derivative gain of the altitude controller, Appendix B (m^3 s/m)
K_m	Gain from volume error to burn rate of the altitude controller, Appendix B (kg s^{-1} m^{-3})
K_p	Proportional gain of the altitude controller, Appendix B (m^3/m)
K_{pre}	Coefficient of the preconditioning rate, Appendix A (kg/s)
L	Net heaviness gained per kg of hydrogen processed
ℓ	Hull length (m)
L_v	Latent heat of water at 25 °C (J/kg)
LHV	Lower heating value of hydrogen (J/kg)
M	Vehicle mass (kg)
M_{str}	Mass of the empty airship (kg)
M_I	Vertical inertia, Equation (2) (kg)
M_{H_2O}, M_{H_2}	Molar mass of water; of hydrogen (kg/mol)
m_{c}	Mass of the hydrogen compressor, in catalog units, Equation (10) (t)
m_{H_2}	Mass of lifting hydrogen in the gas cells (kg)
m_{load}	Payload released at once; in a delivery, the size of each release (kg)
m_{mod}	Mass of a regeneration module, Equation (13) (t)
m_{r}	Hydrogen regenerated per tonne released (kg)
m_{rec}	Lift recovered by regeneration (t)
m_e	Part of the release absorbed by the burn alone with a hold-down device (kg)
m_w	Water retained on board (kg)

$\dot{m}_{H_2}$, $\dot{m}^*_{H_2}$	Hydrogen burn rate; installed rate (kg/s)
$\dot{m}_{\mathrm{PD}}$	Command of the altitude controller, Appendix B (kg/s)
$\dot{m}_{\mathrm{pre}}$	Burn rate that pre-compensates a fraction α of the payload, Appendix A (kg/s)
N	Business cargo delivered over the releases of a trip, Equation (11) (t)
N_{in}	Moles of nitrogen and argon per mole of water in the exhaust
n	Number of consecutive releases of a delivery
P, P_{atm}	Pressure; atmospheric pressure (Pa)
P_c	Heat-rejection power available (W)
P_{cap}	Installed heat-rejection power (W)
P_{rej}	Heat-rejection power needed at the ideal bound (W)
p_{sat}	Saturation pressure of water (Pa)
q, q_0	Heat to reject per kg of hydrogen; with no work exported (J/kg)
Q	Heat to reject per tonne released (GJ/t)
r	Water per kg of hydrogen, M_{H_2O}/M_{H_2}
R_{H_2}	Specific gas constant of hydrogen (J/(kg K))
s	$\sqrt{1 - m_{\mathrm{load}}/M_I}$, Equation (7)
t	Time (s)
t_{anticip}	Anticipation time (s)
t_{burn}	Duration of the clean burn, $m_{\mathrm{load}}/(L\,\dot{m}^*_{H_2})$, Table 3 (s)
t_r	Time from the release to the new equilibrium, Appendix A (s)
t_{rec}	Recovery time with a hold-down device, Equation (8) (s)
T	Time of the consecutive releases of a delivery, Equation (12) (h)
T_{r}	Days of sun allowed to regenerate in flight, Equation (13) (day)
T_1	Temperature of the lifting gas (K)
T_{out}	Exhaust temperature at the exit of the heat-rejection unit (°C)
v	Vertical velocity (m/s)
v_0	Vertical velocity at the release, Appendix A (m/s)
V_{gas}, V_{hull}	Lifting-gas volume; total hull volume (m^3)
V_{eq}	Equilibrium gas volume for the current mass, Appendix B

	(m^3)
$\dot{V}$	Air volume flow into the hull (m^3/s)
W_{comp}	Compression work of the exhaust per kg of hydrogen, Appendix C (J/kg)
α, α^*	Pre-compensated fraction of the payload; value of the sized burn
η	Fraction of the LHV exported as work
λ	Excess-air ratio of a combustor or fuel cell
μ	Force of a hold-down device above the weight of the released load, as a fraction of it
ϕ	Fraction of the product water retained
$\phi_{\max}$	Largest fraction that can condense, Equation (5)
ρ_{air}, ρ_{H_2}	Density of air; of hydrogen (kg/m^3)
ρ_0	Air density at the reference altitude (kg/m^3)
τ	Response time (first-order lag) of the heat-rejection unit (s)
τ_{h}	Response time of a hold-down device, Equation (9) (s)
ω_n	Natural frequency of the hold-down loop in the simulations (rad/s)
ζ	Damping ratio of the altitude loop

References

[1] F. Ben Abdallah, N. Azouz, L. Beji, A. Abichou, Modeling of a heavy-lift airship carrying a payload by a cable-driven parallel manipulator, Int. J. Adv. Robot. Syst. 16 (2019) 1729881419861769. https://doi.org/10.1177/1729881419861769.

[2] G.A. Khoury, J.D. Gillett (Eds.), Airship Technology, Cambridge University Press, Cambridge, UK, 1999.

[3] S.Y. Zhou, M. Renaud, Airship in wood harvesting: the case of the FLYING WHALES' LCA60T, FAO, Rome, 2022. https://www.fao.org/3/cc0997en/cc0997en.pdf (accessed 15 October 2026).

[4] G. Ilieva, J.C. Páscoa, A. Dumas, M. Trancossi, A critical review of propulsion concepts for modern airships, Cent. Eur. J. Eng. 2 (2012) 189–200. https://doi.org/10.2478/s13531-011-0070-1.

[5] E. Lanteigne, A. Alsayed, D. Robillard, S.G. Recoskie, Modeling and control of an unmanned airship with sliding ballast, J. Intell. Robot. Syst. 88 (2017) 285–297. https://doi.org/10.1007/s10846-017-0533-6.

[6] S.A. Barton, Airship buoyancy control using inflatable vacuum chambers, J. Aircr. 50 (2013) 1001–1010. https://doi.org/10.2514/1.C031654.

[7] I. Pasternak, Flight system for a constant volume, variable buoyancy air vehicle (control of static heaviness—COSH), U.S. Patent 9,016,622 B1, 2015. https://patents.google.com/patent/US9016622 (accessed 13 June 2026).

[8] K. Sun, X. Ji, C. Shan, D. Cheng, H. Liang, Extending the flight endurance of stratospheric airships using regenerative fuel cells-assisted pressure maintenance, Renew. Energy 227 (2024) 120470. https://doi.org/10.1016/j.renene.2024.120470.

[9] Goodyear Aerospace Corporation, Feasibility study of modern airships (Phase I), Vol. III: Historical overview (Task I), NASA CR-137692 (Vol. 3), NTIS N75-32034, Akron, OH, 1975. https://ntrs.nasa.gov/citations/19750023961.

[10] W.O.L. Fischer, Ballast water recovery for airships, U.S. Patent 2,078,532, Goodyear-Zeppelin Corporation, 1937. https://patents.google.com/patent/US2078532.

[11] National Aeronautics and Space Administration, Standard for models and simulations, NASA-STD-7009B, Office of the NASA Chief Engineer, Washington, DC, 2024 (approved 5 March 2024). https://standards.nasa.gov/standard/NASA/NASA-STD-7009.

[12] Y. Li, M. Nahon, Modeling and simulation of airship dynamics, J. Guid. Control Dyn. 30 (2007) 1691–1700. https://doi.org/10.2514/1.29061.

[13] X. Wu, C.H. Moog, L.A. Márquez-Martínez, Y. Hu, Full model of a buoyancy-driven airship and its control in the vertical plane, Aerosp. Sci. Technol. 26 (2013) 138–152. https://doi.org/10.1016/j.ast.2012.02.022.

[14] International Organization for Standardization, ISO 2533:1975, Standard Atmosphere, ISO, Geneva, 1975.

[15] NOAA, NASA, USAF, U.S. Standard Atmosphere, 1976, NOAA-S/T 76-1562, U.S. Government Printing Office, Washington, DC, 1976.

[16] H. Lamb, Hydrodynamics, 6th ed., Cambridge University Press, Cambridge, UK, 1932.

[17] L.B. Tuckerman, Inertia factors of ellipsoids for use in airship design, NACA Report No. 210, National Advisory Committee for Aeronautics, Washington, DC, 1926.

[18] Konecranes, Technical information: Konecranes ship-to-shore crane (STS), Hyvänkkä, Finland, 2024. https://www.konecranes.com/system/files/2024-04/7593_sts_monobox_technical_doc_v1.pdf (accessed 8 October 2026).

[19] European Union Aviation Safety Agency, AMC1 SPO.SPEC.HESLO.100 Standard operating procedures, in: AMC/GM to Part-SPO, Amendment 9, Annex to ED Decision 2017/012/R, EASA, Cologne, 2017. https://www.easa.europa.eu/en/downloads/22280/en (accessed 8 October 2026).

[20] R.W. Knight, The Hindenburg accident: a comparative digest of the investigations and findings, with the American and translated German reports included, Report No. 11, Safety and Planning Division, Bureau of Air Commerce, U.S. Department of Commerce, Washington, DC, 1938. https://doi.org/10.21949/1511799.

[21] L. Dürr, The American airship ZR-3, NACA Technical Memorandum No. 286, National Advisory Committee for Aeronautics, Washington, DC, 1924. https://ntrs.nasa.gov/citations/19930086540.

[22] Flying Whales, The LCA60T and Frequently asked questions. https://www.flying-whales.com/en/the-lca60t/ and https://www.flying-whales.com/en/faq/ (accessed 28 September 2026).

[23] European Union Aviation Safety Agency, Type-Certificate Data Sheet No. EASA.AS.001, Zeppelin LZ N07-100 and LZ N07-101, Issue 10.

https://www.easa.europa.eu/en/document-library/type-certificates/airship-cs-30-cs-31ha/easaas001-zeppelin-lz-n07-100.

[24] International Association for the Properties of Water and Steam, Revised release on the IAPWS Industrial Formulation 1997 for the thermodynamic properties of water and steam, IAPWS R7-97(2012), 2012.

[25] G. Lopez-Ruiz, J. Castresana-Larrauri, J.M. Blanco-Ilzarbe, Thermodynamic analysis of a regenerative Brayton cycle using H_2, CH_4 and H_2/CH_4 blends as fuel, Energies 15 (2022) 1508. https://doi.org/10.3390/en15041508.

[26] C.M. Ivler, J.D. Powell, M.B. Tischler, J.W. Fletcher, C. Ott, Design and flight test of a cable angle/rate feedback flight control system for the RASCAL JUH-60 helicopter, in: American Helicopter Society 68th Annual Forum, Fort Worth, TX, 2012.

[27] P. Lambrechts, M. Boerlage, M. Steinbuch, Trajectory planning and feedforward design for high performance motion systems, in: Proceedings of the 2004 American Control Conference, Vol. 5, Boston, MA, 2004, pp. 4637–4642. https://doi.org/10.23919/ACC.2004.1384042.

[28] S. Verhelst, T. Wallner, Hydrogen-fueled internal combustion engines, Prog. Energy Combust. Sci. 35 (2009) 490–527. https://doi.org/10.1016/j.pecs.2009.08.001.

[29] L. Zhang, S. Spatari, Y. Sun, Life cycle assessment of novel heat exchanger for dry cooling of power plants based on encapsulated phase change materials, Appl. Energy 271 (2020) 115227, with supplementary material. https://doi.org/10.1016/j.apenergy.2020.115227.

[30] BRP-Rotax GmbH & Co KG, Rotax 916 iS/c, product folder, Gunskirchen, Austria, 2023. https://www.flyrotax.com/assets/files/products/Rotax-916iSc-Product_Folder.pdf (accessed 27 September 2026).

[31] DEUTZ AG, DEUTZ TCG 7.8 H_2, product datasheet, Cologne, 2024. https://www.deutz.com.au/wp-content/uploads/2025/05/DEUTZ_ProductDatasheet_TCG_7.8_H2.pdf (accessed 27 September 2026).

[32] K. Simmons, Z. Hulvey, A.-D. Celestine, Onboard Type IV compressed hydrogen storage system—cost and performance status, DOE Hydrogen Program Record 24006, U.S. Department of Energy, Washington, DC, 2025. https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/24006-onboard-storage-cost-performance-status.pdf (accessed 29 September 2026).

[33] Hydro-Pac Inc., LX-Series hydrogen gas compressors, 1500 to 12000 psi, technical brochure, Fairview, PA, 2011. https://www.hydropac.com/literature/LX-SERIES%20Hydrogen%20Brochure%2001_2011-%20reduced.pdf (accessed 27 September 2026).

[34] X. Zhang, H. Yue, H. Zheng, L. Tan, Z. Zhang, F. Li, Proton exchange membrane fuel cells for aircraft applications: a comprehensive review of key challenges and development trends, Hydrogen 6 (2025) 116. https://doi.org/10.3390/hydrogen6040116.

[35] U.S. Department of Defense, MIL-T-53029C, Military specification: tanks, fabric, collapsible: 3,000, 10,000, 20,000, and 50,000 gallon, drinking water, 11 May 1994. https://everyspec.com/MIL-SPECS/MIL-SPECS-MIL-T/MIL-T-53029C_51899/ (accessed 13 October 2026).

[36] Siemens AG, Gas turbine SGT-800 for power generation applications, data sheet, Article No. PGDG-T10040-00-7600, Power and Gas Division, Erlangen, Germany, 2017. https://assets.new.siemens.com/siemens/assets/api/uuid:daeb2935-2ea5-4400-a5dc-653de64ca7d3/gas-turbine-sgt-800.pdf (accessed 11 October 2026).

[37] T. Pattanayak, D. Mavris, Battery technology for sustainable aviation: a review of current trends and future prospects, Appl. Energy 397 (2025) 126356. https://doi.org/10.1016/j.apenergy.2025.126356.

[38] Aerospace Technology Institute, FlyZero: Thermal Management Roadmap Report, FZO-PPN-COM-0019, ATI, Cranfield, UK, 2022. https://www.ati.org.uk/wp-content/uploads/2022/03/FZO-PPN-COM-0019-Thermal-Management-Roadmap-Report.pdf (accessed 24 September 2026).

[39] P. Ziegler, C. Simson, T. Arzberger, A. Woitalka, F. Nöske, S. Kaiser, V. Gümmer, Water-Enhanced Turbofan condenser sizing considering

configuration and operating conditions, Aeronaut. J. 129 (2025) 1494–1514. https://doi.org/10.1017/aer.2024.158.

[40] S. Bougon, B. Le François, Device and method for ballast management, European Patent EP 3980329 B1, Flying Whales SAS, 2024. https://patents.google.com/patent/EP3980329B1 (accessed 15 October 2026).

[41] J. Adams, C. Houchins, R. Ahluwalia, Onboard Type IV compressed hydrogen storage system—cost and performance status, DOE Hydrogen and Fuel Cells Program Record 19008, U.S. Department of Energy, Washington, DC, 2019. https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/19008_onboard_storage_cost_performance_status.pdf (accessed 29 September 2026).

[42] International Energy Agency, Global Hydrogen Review 2025, Assumptions annex, IEA, Paris, 2025.

[43] M. Gardiner, Energy requirements for hydrogen gas compression and liquefaction as related to vehicle storage needs, DOE Hydrogen and Fuel Cells Program Record 9013, U.S. Department of Energy, Washington, DC, 2009. https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/9013_energy_requirements_for_hydrogen_gas_compression.pdf (accessed 27 September 2026).

[44] A. Giuffrida, P. Colbertaldo, Overview of diaphragm compressors for hydrogen service: capacity, discharge pressure and operational challenges, J. Energy Storage 129 (2025) 117286. https://doi.org/10.1016/j.est.2025.117286.

[45] K. Sun, Y. Jiang, H. Liu, Configuration optimization of PV array for stratospheric airship under nonuniform radiation condition, Aerosp. Sci. Technol. 155 (2024) 109550. https://doi.org/10.1016/j.ast.2024.109550.

[46] SENAMHI, MINEM, Atlas de Energía Solar del Perú, SENAMHI, Lima, Peru, 2003. https://www.senamhi.gob.pe/pdf/Atlas%20_de_Radiacion_Solar.pdf (accessed 13 June 2026).

[47] J.A. Duffie, W.A. Beckman, Solar Engineering of Thermal Processes, 4th ed., Wiley, Hoboken, NJ, 2013. https://doi.org/10.1002/9781118671603.

[48] J.K. Page, The estimation of monthly mean values of daily total short wave radiation on vertical and inclined surfaces from sunshine records for latitudes 40°N–40°S, in: Proceedings of the United Nations Conference on New Sources of Energy, Rome, 21–31 August 1961, vol. 4, Solar Energy I, United Nations, New York, 1964, pp. 378–390. https://digitallibrary.un.org/record/3824017 (accessed 27 September 2026).

[49] D. Pande, D. Verstraete, Impact of solar cell characteristics and operating conditions on the sizing of a solar powered nonrigid airship, Aerosp. Sci. Technol. 72 (2018) 353–363. https://doi.org/10.1016/j.ast.2017.11.013.

[50] MiaSolé, CIGS thin-film efficiency: flexible modules (FLEX series product brief), MiaSolé Hi-Tech Corp., Santa Clara, CA, 2014. https://miasole.com/wp-content/uploads/2014/09/FLEX_Benefits.pdf (accessed 25 September 2026).

[51] E.W. Lemmon, M.O. McLinden, D.G. Friend, Thermophysical properties of fluid systems, in: P.J. Linstrom, W.G. Mallard (Eds.), NIST Chemistry WebBook, NIST Standard Reference Database 69, National Institute of Standards and Technology, Gaithersburg, MD. https://doi.org/10.18434/T4D303 (accessed 24 September 2026).

[52] J.W. Leachman, R.T. Jacobsen, S.G. Penoncello, E.W. Lemmon, Fundamental equations of state for parahydrogen, normal hydrogen, and orthohydrogen, J. Phys. Chem. Ref. Data 38 (2009) 721–748. https://doi.org/10.1063/1.3160306.

[53] L. Zakhvatkin, A. Schechter, I. Avrahami, Water recuperation from hydrogen fuel cell during aerial mission, Energies 15 (2022) 6848. https://doi.org/10.3390/en15186848.

[54] T. Yamasaki, N. Goto, Identification of blimp dynamics via flight tests, Trans. Jpn. Soc. Aeronaut. Space Sci. 46 (2003) 195–205. https://doi.org/10.2322/tjsass.46.195.

[55] S.W. Churchill, H.H.S. Chu, Correlating equations for laminar and turbulent free convection from a vertical plate, Int. J. Heat Mass Transfer 18 (1975) 1323–1329. https://doi.org/10.1016/0017-9310(75)90243-4.

[56] Transport Canada, Airworthiness Manual Chapter 541 – Airships, Canadian Aviation Regulations, Change 541-2, Ottawa, 2009, Section 541.7. https://tc.canada.ca/en/corporate-services/acts-regulations/list-regulations/canadian-aviation-regulations-sor-96-433/standards/airworthiness-chapter-541-airships-canadian-aviation-regulations-cars (accessed 13 October 2026).

[57] SAE International, ARP4761A: Guidelines for conducting the safety assessment process on civil aircraft, systems, and equipment, SAE International, Warrendale, PA, 2023. https://saemobilus.sae.org/standards/arp4761a-guidelines-conducting-safety-assessment-process-civil-aircraft-systems-equipment.

[58] D. Howe, Ground handling and mooring, in: G.A. Khoury, J.D. Gillett (Eds.), Airship Technology, Cambridge University Press, Cambridge, UK, 1999, pp. 297–320.